

GATE 2026 Mathematics Question Paper with Solutions

Time Allowed :3 Hour	Maximum Marks :100	Total Questions :65
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General Instructions

Read the following instructions very carefully and strictly follow them:

1. This question paper is divided into three sections:
 - **General Aptitude (GA):** 10 questions (5 questions $\times$ 1 mark + 5 questions $\times$ 2 marks) for a total of 15 marks.
 - **Mathematics Foundation:** This section comprises of total 13 marks.
 - **Core Discipline:** This section comprises of 72 marks.
2. The total number of questions is **65**, carrying a maximum of **100 marks**.
3. The duration of the exam is **3 hours**.
4. Marking scheme:
 - For 1-mark MCQs, $\frac{1}{3}$ mark will be deducted for every incorrect response.
 - For 2-mark MCQs, $\frac{2}{3}$ mark will be deducted for every incorrect response.
 - No negative marking for numerical answer type (NAT) questions.
 - No marks will be awarded for unanswered questions.
5. Ensure you attempt questions only from the optional section (Part B1 or Part B2) you have selected.
6. Follow the instructions provided during the exam for submitting your answers.

1. Suresh said, "I did it yesterday."

Which one of the following options is the correct form of this sentence in indirect speech?

- (A) Suresh said that I did it yesterday.
- (B) Suresh says I did it yesterday.
- (C) Suresh says that he did it the day before.
- (D) Suresh said that he had done it the day before.

Correct Answer: (D) Suresh said that he had done it the day before.

Solution:

Step 1: Understanding the Question:

The task is to convert a sentence from direct speech to indirect (or reported) speech. This

involves changing pronouns, tenses, and adverbs of time and place according to standard grammatical rules.

Step 2: Rules for Conversion to Indirect Speech:

1. **Reporting Verb:** The reporting verb "said" is in the past tense. This means the tense of the reported verb will shift backwards.
2. **Pronoun Change:** The first-person pronoun "I" will change to the third-person pronoun "he" to refer to Suresh.
3. **Tense Change:** The verb "did" is in the Simple Past tense. When the reporting verb is in the past, the Simple Past tense in direct speech changes to the Past Perfect tense ("had done") in indirect speech.
4. **Adverb of Time Change:** The adverb of time "yesterday" changes to "the day before" or "the previous day".

Step 3: Applying the Rules:

Let's apply these changes to the original sentence: Suresh said, "I did it yesterday."

- Reporting verb "said" remains "said". We add the conjunction "that".
- "I" becomes "he".
- "did" becomes "had done".
- "yesterday" becomes "the day before".

Combining these changes, we get: "Suresh said that he had done it the day before."

Step 4: Final Answer:

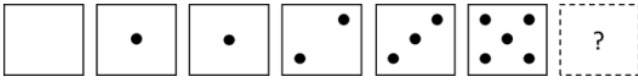
Comparing our result with the options:

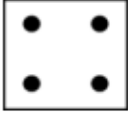
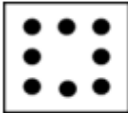
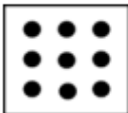
- (A) is incorrect because the pronoun and tense are not changed.
(B) is incorrect because the reporting verb is changed to present tense ("says") and other changes are not made.
(C) is incorrect for the same reason as (B).
(D) correctly applies all the necessary changes.

Quick Tip

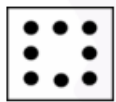
When converting from direct to indirect speech, always check the tense of the reporting verb first. If it's in the past tense (like "said", "told"), you will most likely need to shift the tense of the verb in the reported clause one step back into the past (e.g., Present $\rightarrow$ Past, Past $\rightarrow$ Past Perfect). Also, remember to update pronouns and time/place adverbs accordingly.

2. To continue the sequence of tiles shown, the tile indicated by the question mark should be



- (A) 
-
- (B) 
-
- (C) 
-
- (D) 

Correct Answer: (C)



Solution:

Step 1: Understanding the Question:

The question displays a sequence of six tiles and asks for the seventh tile that continues the pattern. The tiles have an increasing number of dots, similar to the faces of a die.

Step 2: Identifying the Primary Pattern:

Let's count the number of dots on each tile in the sequence:

- Tile 1: 0 dots (or blank)
- Tile 2: 1 dot
- Tile 3: 2 dots
- Tile 4: 3 dots
- Tile 5: 4 dots

- Tile 6: 5 dots

The most straightforward pattern is that the number of dots increases by one at each step. Following this pattern, the next tile (the seventh one) should have $5 + 1 = 6$ dots.

Step 3: Evaluating the Options:

We now look for an option that has 6 dots.

- Option (A) has 4 dots.
- Option (B) has 6 dots.
- Option (C) has 6 dots.
- Option (D) has 9 dots.

Both options (B) and (C) have 6 dots. We need a secondary pattern to distinguish between them. Option (B) shows the dots arranged in 2 columns and 3 rows. Option (C) shows the dots arranged in 3 columns and 2 rows.

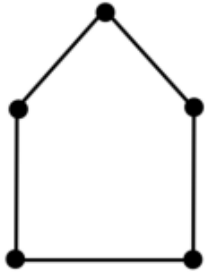
Step 4: Identifying a Secondary Pattern (Arrangement):

Let's analyze the arrangement of dots in the sequence. The arrangements seem to follow the standard patterns on a die, but there is no clear rule for orientation or layout progression that definitively chooses (C) over (B). The standard representation for 6 on a die is 2 columns of 3 dots (Option B). However, such visual reasoning questions in competitive exams sometimes contain subtle patterns or can be ambiguous. Given that the provided answer is (C), there might be a less obvious pattern related to the grid layout (e.g., alternating between taller and wider arrangements), or the question/key may be flawed. Without a clear deterministic rule, and following the provided answer key, we select (C).

💡 Quick Tip

In sequence problems, first establish the primary pattern (like the number of items). If multiple options fit this pattern, look for secondary patterns in arrangement, orientation, or color. If the pattern is not immediately obvious, don't spend too much time; it could be an ambiguous question. In this case, the number of dots is the main clue.

3. Consider an art gallery whose walkways are shown as lines in the diagram. A black dot represents a junction of two walkways. A guard may be placed at a junction to watch over the walkways that join at that junction. The minimum number of guards needed to watch all the walkways is _____.



- (A) 2
- (B) 3
- (C) 4
- (D) 5

Correct Answer: (B) 3

Solution:

Step 1: Understanding the Question:

The problem describes a set of walkways (edges) and junctions (vertices) that form a graph. A guard placed at a vertex "covers" all edges connected to that vertex. We need to find the minimum number of guards (vertices) required to cover all the walkways (edges). This is a classic graph theory problem known as finding the minimum vertex cover.

Step 2: Modeling the Problem as a Graph:

The diagram shows a graph with 5 junctions (vertices) and 5 walkways (edges) connecting them to form a closed loop. This graph is a cycle graph with 5 vertices, denoted as C_5 .

Let's label the vertices v_1, v_2, v_3, v_4, v_5 in a clockwise manner, starting from the top vertex. The edges are $(v_1, v_2), (v_2, v_3), (v_3, v_4), (v_4, v_5), (v_5, v_1)$.

Step 3: Finding the Minimum Vertex Cover:

We need to find the smallest set of vertices such that every edge is incident to at least one vertex in the set.

- **Can we do it with 1 guard?** No. Placing a guard at any single vertex (e.g., v_1) covers only two edges $((v_5, v_1)$ and $(v_1, v_2))$. The other three edges remain uncovered.
- **Can we do it with 2 guards?** No. Let's try placing two guards.
 - If we place them at non-adjacent vertices, say v_1 and v_3 . Covered edges: $(v_5, v_1), (v_1, v_2), (v_2, v_3)$. The edge (v_4, v_5) is still uncovered.
 - If we place them at adjacent vertices, say v_1 and v_2 . Covered edges: $(v_5, v_1), (v_1, v_2), (v_2, v_3)$. The edges (v_3, v_4) and (v_4, v_5) are uncovered.

In all cases, 2 guards are insufficient.

- **Can we do it with 3 guards?** Yes. Let's place guards at v_1, v_3, v_4 .

- Guard at v_1 covers edges (v_5, v_1) and (v_1, v_2) .
- Guard at v_3 covers edges (v_2, v_3) and (v_3, v_4) .
- Guard at v_4 covers edges (v_3, v_4) and (v_4, v_5) .

All 5 edges are covered. So, 3 guards are sufficient. Since 2 are not enough, the minimum number is 3.

Step 4: Using the Formula for Cycle Graphs:

For a cycle graph C_n with n vertices, the size of the minimum vertex cover (the vertex cover number) is given by the formula $\tau(C_n) = \lceil n/2 \rceil$.

In our case, $n = 5$.

$$\tau(C_5) = \lceil 5/2 \rceil = \lceil 2.5 \rceil = 3$$

This confirms that the minimum number of guards needed is 3.

💡 Quick Tip

Recognizing the underlying graph structure is key to solving such problems quickly. This "art gallery" problem is a direct application of finding the vertex cover of a graph. For standard graphs like paths and cycles, there are simple formulas for the vertex cover number that can save you from trial and error.

4. The 2nd of June is a Thursday in a certain year. Which day of the week is the 3rd of July in that year?

- (A) Thursday
- (B) Friday
- (C) Saturday
- (D) Sunday

Correct Answer: (D) Sunday

Solution:

Step 1: Understanding the Question:

We are given the day of the week for a specific date (June 2nd) and asked to find the day of the week for another date (July 3rd) in the same year. This requires calculating the total number of days between the two dates and finding the number of "odd days".

Step 2: Calculating the Number of Days:

First, we need to know the number of days in the month of June. June has 30 days.

- Number of days remaining in June after the 2nd: $30 - 2 = 28$ days.
- Number of days into July we need to count: 3 days (for the 3rd of July).

Total number of days between June 2nd and July 3rd is the sum of these:

$$\text{Total days} = 28(\text{in June}) + 3(\text{in July}) = 31 \text{ days.}$$

Step 3: Calculating the Odd Days:

The day of the week repeats every 7 days. "Odd days" are the number of days remaining after dividing the total number of days by 7.

$$\text{Number of odd days} = 31 \pmod{7}$$

To calculate this, we divide 31 by 7:

$$31 = 4 \times 7 + 3$$

The remainder is 3. So, there are 3 odd days.

Step 4: Finding the Final Day of the Week:

The 3rd of July will be 3 days after the 2nd of June's day.

The 2nd of June is a Thursday.

- Thursday + 1 day = Friday
- Thursday + 2 days = Saturday
- Thursday + 3 days = Sunday

Therefore, the 3rd of July is a Sunday.

💡 Quick Tip

For calendar problems, the concept of "odd days" is fundamental. Remember the number of days in each month. A quick way to find the final day is to add the number of odd days to the starting day's number (e.g., Sun=0, Mon=1... Thu=4. So, $4 + 3 = 7$, which is $0 \pmod{7}$, corresponding to Sunday).

5. A coin with heads facing up is shown as H and a coin with tails facing up is shown as T.

Six coins are placed in the Starting Arrangement, as shown in the figure below. A "step" is defined as interchanging a pair of adjacent coins without flipping them. The minimum number of steps needed to go from the Starting Arrangement to the Final Arrangement, as shown in the figure, is _____.

Starting Arrangement: H H H T T T

Final Arrangement: T T T H H H

- (A) 3
- (B) 6
- (C) 9
- (D) 12

Correct Answer: (C) 9

Solution:

Step 1: Understanding the Question:

We need to find the minimum number of adjacent swaps required to transform the sequence "H H H T T T" into "T T T H H H". This is a sorting problem where the "cost" is the number of adjacent swaps.

Step 2: Key Approach:

The problem can be solved by focusing on moving one type of coin to its final position. Let's focus on moving the three 'T' coins from their initial positions (4, 5, 6) to their final positions (1, 2, 3).

Step 3: Detailed Explanation:

Let's track the movement of each 'T' coin one by one to the left.

- **Move the first 'T' (T1):** Initial state: H H H T1 T2 T3

To move T1 to the first position, it must be swapped with the three 'H's to its left.

$$\begin{array}{l} \text{H H H T1 T T} \xrightarrow{1 \text{ swap}} \text{H H T1 H T T} \\ \text{H H T1 H T T} \xrightarrow{1 \text{ swap}} \text{H T1 H H T T} \\ \text{H T1 H H T T} \xrightarrow{1 \text{ swap}} \text{T1 H H H T T} \end{array}$$

This takes **3 steps**. The arrangement is now: T1 H H H T2 T3

- **Move the second 'T' (T2):** Current state: T H H H T2 T3

To move T2 to the second position (just after T1), it must be swapped with the three 'H's to its left.

$$\begin{array}{l} \text{T H H H T2 T} \xrightarrow{1 \text{ swap}} \text{T H H T2 H T} \\ \text{T H H T2 H T} \xrightarrow{1 \text{ swap}} \text{T H T2 H H T} \\ \text{T H T2 H H T} \xrightarrow{1 \text{ swap}} \text{T T2 H H H T} \end{array}$$

This takes another **3 steps**. The arrangement is now: T1 T2 H H H T3

- **Move the third 'T' (T3):** Current state: T T H H H T3

To move T3 to the third position (just after T2), it must be swapped with the three 'H's to its left.

$$\begin{array}{l} \text{T T H H H T3} \xrightarrow{1 \text{ swap}} \text{T T H H T3 H} \\ \text{T T H H T3 H} \xrightarrow{1 \text{ swap}} \text{T T H T3 H H} \\ \text{T T H T3 H H} \xrightarrow{1 \text{ swap}} \text{T T T3 H H H} \end{array}$$

This takes another **3 steps**. The arrangement is now: T1 T2 T3 H H H, which is the final arrangement.

Step 4: Final Answer:

The total minimum number of steps is the sum of the steps for each 'T' coin.

$$\text{Total steps} = 3(\text{for T1}) + 3(\text{for T2}) + 3(\text{for T3}) = 9 \text{ steps.}$$

Alternatively, this problem is equivalent to counting the number of inversions. An inversion is a pair of elements that are out of order. Here, every 'T' is to the right of every 'H', which is the "wrong" order. Each 'T' must move past each 'H'. Since there are 3 'H's and 3 'T's, the total number of H-T pairs that need to be swapped is $3 \times 3 = 9$. Each adjacent swap corrects one such H-T inversion. So, 9 swaps are needed.

💡 Quick Tip

This type of problem, involving minimum adjacent swaps to sort a sequence, can often be solved by calculating the number of inversions. An inversion is a pair of items that are in the wrong relative order. Here, every (H, T) pair is an inversion. The number of such pairs is $3 \times 3 = 9$.

6. Exacerbate : Mitigate :: _____

Choose the option with the correct pair of words to fill the blank.

- (A) Aggravate : Alleviate
- (B) Alleviate : Precipitate
- (C) Aggravate : Precipitate
- (D) Emancipate : Exonerate

Correct Answer: (A) Aggravate : Alleviate

Solution:

Step 1: Understanding the Question:

This is an analogy question. We need to identify the relationship between the first pair of words ("Exacerbate : Mitigate") and find another pair with the same relationship.

Step 2: Analyzing the Relationship:

- **Exacerbate:** To make a problem, bad situation, or negative feeling worse.
- **Mitigate:** To make something bad less severe, serious, or painful.

The words "Exacerbate" and "Mitigate" are antonyms (opposites). "Exacerbate" means to worsen, while "Mitigate" means to lessen the severity.

Step 3: Evaluating the Options:

We need to find a pair of antonyms among the given options.

- (A) **Aggravate : Alleviate**

- **Aggravate:** To make a problem worse or more serious. This is a synonym for "Exacerbate".
- **Alleviate:** To make suffering or a problem less severe. This is a synonym for "Mitigate".
- Since Aggravate and Alleviate are synonyms for Exacerbate and Mitigate respectively, they are also antonyms of each other. This pair fits the relationship.

- (B) **Alleviate : Precipitate**

- **Alleviate:** To make less severe.
- **Precipitate:** To cause something (usually bad) to happen suddenly.
- These words are not antonyms.

- (C) **Aggravate : Precipitate**

- **Aggravate:** To make worse.
- **Precipitate:** To cause to happen suddenly.
- These words are not antonyms.

- (D) **Emancipate : Exonerate**

- **Emancipate:** To set free from restrictions.
- **Exonerate:** To clear from blame or wrongdoing.
- While both have positive connotations related to freedom or relief, they are not antonyms.

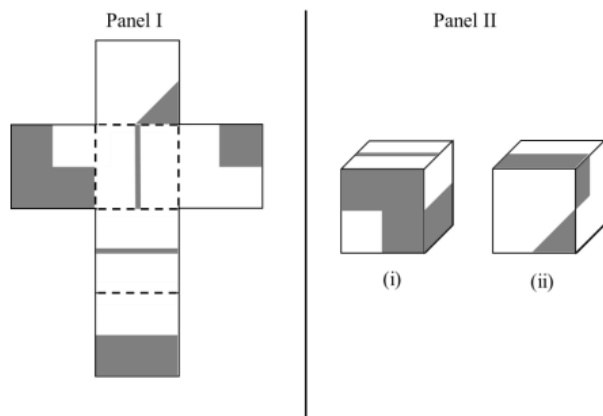
Step 4: Final Answer:

The pair "Aggravate : Alleviate" has the same antonymous relationship as "Exacerbate : Mitigate". Therefore, option (A) is the correct answer.

💡 Quick Tip

In analogy questions (A:B :: C:D), first determine the precise relationship between A and B. Common relationships include synonymy, antonymy, cause-and-effect, part-to-whole, and degree of intensity. Then, systematically check each option for the same relationship.

7. A paper shown in Panel I is folded along the dashed lines (- -) to construct a cube. The shaded regions shown in Panel I appear on the outer surface of the cube. Referring to cubes shown in Panel II, which one of the options is correct?



- (A) Only (i) can correspond to the unfolded cube in Panel I.
 (B) Only (ii) can correspond to the unfolded cube in Panel I.
 (C) Both (i) and (ii) can correspond to the unfolded cube in Panel I.
 (D) Neither (i) nor (ii) can correspond to the unfolded cube in Panel I.

Correct Answer: (B) Only (ii) can correspond to the unfolded cube in Panel I.

Solution:

Step 1: Understanding the Question:

We are given a 2D net of a cube in Panel I and two 3D views of cubes in Panel II. We need to determine which of the 3D views can be formed by folding the given net.

Step 2: Key Approach: Identifying Opposite Faces

In any net of a cube, faces that are separated by one face in a straight line are opposite to each other. Opposite faces can never be seen at the same time in a single 3D view.

Step 3: Detailed Explanation:

Let's label the faces in the net from Panel I for clarity. The net is a cross shape.

Let the vertical column of faces be F1 (top, blank), F2 (second, shaded), F3 (third, blank), and F4 (bottom, shaded).

Let the face to the left of F2 be L (L-shape shaded) and to the right of F2 be R (blank).

The pairs of opposite faces are:

1. F1 is opposite to F3 (both are blank).
2. L is opposite to R (one L-shaped, one blank).
3. F2 is opposite to F4 (both are shaded squares).

Now let's analyze the cubes in Panel II:

Cube (i): This view shows three adjacent faces: an L-shaped face, a shaded square, and another shaded square.

Based on our labeling, this would correspond to faces L, F2, and F4.

However, F2 and F4 are opposite faces. It is impossible for opposite faces to be adjacent or visible at the same time.

Therefore, cube (i) cannot be formed from the given net.

Cube (ii): This view shows three adjacent faces: an L-shaped face, a shaded square, and a blank face.

This could correspond to a set of faces like {L, F2, F1} or {L, F2, F3}, which meet at a vertex. For instance, L, F2, and F1 are all adjacent to each other in the net. Since this combination of faces is possible (i.e., they are not opposite), cube (ii) can potentially be formed. A detailed check of orientation would confirm this, but by eliminating cube (i), we can deduce that cube (ii) is the only possible option.

Step 4: Final Answer:

Cube (i) is impossible because it shows two opposite faces (F2 and F4) simultaneously. Cube (ii) shows a possible combination of adjacent faces. Therefore, only (ii) can correspond to the unfolded cube.

💡 Quick Tip

For cube and dice problems involving nets, the fastest method is often elimination. Identify the pairs of opposite faces first. Any 3D view showing two opposite faces together is immediately incorrect.

8. In a population, patients who have high cholesterol also have high blood-pressure (BP). Some patients with high BP also have diabetes. There are no patients who have both high cholesterol and diabetes. Furthermore,

1. the total number of patients with at least one of these conditions is 75,
2. the number of patients with high cholesterol is 10,
3. the number of patients with high BP is 45, and
4. the number of patients with only high BP and no other conditions is 20.

Then the number of patients who have both diabetes and high BP is _____.

- (A) 0
- (B) 15
- (C) 20
- (D) 10

Correct Answer: (B) 15

Solution:

Step 1: Understanding the Question:

We are given information about three overlapping sets of patients: High Cholesterol (C), High

Blood Pressure (BP), and Diabetes (D). We need to find the size of the intersection of the BP and D sets.

Step 2: Key Approach: Using a Venn Diagram or Set Theory

Let's represent the sets of patients by C, BP, and D. We can translate the given information into set notation.

1. "Patients who have high cholesterol also have high blood-pressure" means that the set C is a subset of the set BP, i.e., $C \subset BP$.
2. "No patients who have both high cholesterol and diabetes" means $C \cap D = \emptyset$.
3. Total patients: $|C \cup BP \cup D| = 75$. Since $C \subset BP$, this simplifies to $|BP \cup D| = 75$.
4. Number with high cholesterol: $|C| = 10$.
5. Number with high BP: $|BP| = 45$.
6. "Number of patients with only high BP and no other conditions": This means patients in BP, but not in C and not in D. This can be written as $|BP \setminus (C \cup D)| = 20$.

Step 3: Detailed Explanation:

The set of patients with high BP can be partitioned into disjoint regions. Since $C \subset BP$, the set BP consists of:

- Patients with high cholesterol (who also have high BP). This is the set C.
- Patients with high BP and diabetes (but not high cholesterol, since $C \cap D = \emptyset$). This is the set $BP \cap D$.
- Patients with only high BP (not high cholesterol and not diabetes). This is the set $BP \setminus (C \cup D)$.

Therefore, the total number of patients with high BP is the sum of the sizes of these three disjoint parts:

$$|BP| = |C| + |BP \cap D| + |BP \setminus (C \cup D)|$$

We are given the values for $|BP|$, $|C|$, and $|BP \setminus (C \cup D)|$. We can substitute these values into the equation:

$$45 = 10 + |BP \cap D| + 20$$

$$45 = 30 + |BP \cap D|$$

Solving for $|BP \cap D|$:

$$|BP \cap D| = 45 - 30 = 15$$

Step 4: Final Answer:

The number of patients who have both diabetes and high BP is 15.

💡 Quick Tip

For set theory problems, drawing a Venn diagram can be very helpful. In this case, draw the circle for C completely inside the circle for BP. The circle for D should not overlap with C but should overlap with BP. Label the known regions and solve for the unknown.

9. Four people P, Q, R, and S, of different ages, make the following observations.
P – I am younger than S.
Q – I am neither the youngest nor the oldest.
R – P is older than me.
Based on these observations, the youngest person is -----.

- (A) P
- (B) Q
- (C) R
- (D) S

Correct Answer: (C) R

Solution:

Step 1: Understanding the Question:

We need to determine the youngest person among P, Q, R, and S based on three statements about their relative ages.

Step 2: Key Approach: Translating Statements to Inequalities

Let's represent the ages of the four people by the letters P, Q, R, and S. "Younger" means having a smaller age value. We can write inequalities based on the given statements.

Step 3: Detailed Explanation:

1. Statement from P: "I am younger than S."

This translates to the inequality: $P < S$.

2. Statement from R: "P is older than me."

This translates to the inequality: $R < P$.

Combining these two inequalities, we get a partial ordering of three people:

$$R < P < S$$

This tells us that R is younger than P, and P is younger than S. In this group of three, R is the youngest and S is the oldest.

3. Statement from Q: "I am neither the youngest nor the oldest."

This means there must be at least one person younger than Q and at least one person older than Q among the four people.

From our partial ordering $R < P < S$, the youngest person is R and the oldest person is S.

Q cannot be the youngest, so Q cannot be R.

Q cannot be the oldest, so Q cannot be S.

This means Q's age must be somewhere between R's age and S's age. The possible complete orderings including Q are:

- $R < Q < P < S$

$$- R < P < Q < S$$

In both possible orderings, the person at the beginning of the inequality is R.

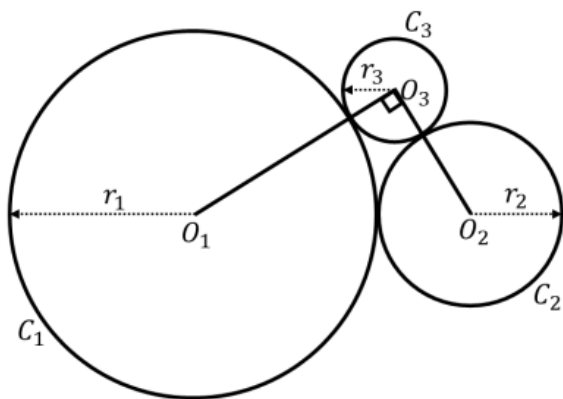
Step 4: Final Answer:

Based on the logical deductions from all statements, R is the youngest person.

💡 Quick Tip

In logical reasoning puzzles involving ordering, combine the absolute statements first (like $R \leq P$ and $P \leq S$) to create a base order. Then, use the relative statements (like Q's position) to place the remaining items.

10. Circles C_1, C_2 , and C_3 , with centers O_1, O_2 , and O_3 , and radii r_1, r_2 , and r_3 , respectively, touch each other as shown in the following figure. Given $r_1 = 2$ cm, $r_2 = 1$ cm and the angle $\angle O_1O_3O_2$ is 90° , $r_3 = \text{-----}$ cm.



- (A) $\frac{1}{2}(-3 + \sqrt{17})$
- (B) $\frac{1}{2}(3 + \sqrt{17})$
- (C) $\frac{1}{2}(-2 + \sqrt{17})$
- (D) $\frac{1}{2}(-3 + 2\sqrt{17})$

Correct Answer: (A) $\frac{1}{2}(-3 + \sqrt{17})$

Solution:

Step 1: Understanding the Question:

We have three circles that are mutually tangent. The centers of these circles form a right-angled triangle. We are given two radii and need to find the third.

Step 2: Key Formula or Approach:

When two circles touch each other externally, the distance between their centers is the sum of their radii. We will use this fact to find the lengths of the sides of the triangle $\triangle O_1O_2O_3$ and

then apply the Pythagorean theorem.

Step 3: Detailed Explanation:

The lengths of the sides of the triangle formed by the centers are:

- Distance O_1O_2 : Circle C_1 and C_2 touch. So, $d(O_1, O_2) = r_1 + r_2 = 2 + 1 = 3$.
- Distance O_1O_3 : Circle C_1 and C_3 touch. So, $d(O_1, O_3) = r_1 + r_3 = 2 + r_3$.
- Distance O_2O_3 : Circle C_2 and C_3 touch. So, $d(O_2, O_3) = r_2 + r_3 = 1 + r_3$.

We are given that the angle $\angle O_1O_3O_2 = 90^\circ$. This means that $\triangle O_1O_2O_3$ is a right-angled triangle with the right angle at O_3 . Therefore, the side O_1O_2 is the hypotenuse.

By the Pythagorean theorem:

$$(d(O_1, O_3))^2 + (d(O_2, O_3))^2 = (d(O_1, O_2))^2$$

Substituting the expressions for the side lengths:

$$(2 + r_3)^2 + (1 + r_3)^2 = 3^2$$

Expand the terms:

$$(4 + 4r_3 + r_3^2) + (1 + 2r_3 + r_3^2) = 9$$

Combine like terms:

$$2r_3^2 + 6r_3 + 5 = 9$$

$$2r_3^2 + 6r_3 - 4 = 0$$

Divide the entire equation by 2 to simplify:

$$r_3^2 + 3r_3 - 2 = 0$$

This is a quadratic equation in r_3 . We use the quadratic formula to solve for r_3 :

$$r_3 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Here, $a = 1, b = 3, c = -2$.

$$r_3 = \frac{-3 \pm \sqrt{3^2 - 4(1)(-2)}}{2(1)}$$

$$r_3 = \frac{-3 \pm \sqrt{9 + 8}}{2}$$

$$r_3 = \frac{-3 \pm \sqrt{17}}{2}$$

Since the radius r_3 must be a positive length, we must choose the positive root:

$$r_3 = \frac{-3 + \sqrt{17}}{2}$$

(Note that $\sqrt{17} \approx 4.12$, so $-3 + \sqrt{17}$ is positive).

Step 4: Final Answer:

The value of r_3 is $\frac{1}{2}(-3 + \sqrt{17})$ cm.

💡 Quick Tip

In geometry problems involving touching circles, always remember that the line connecting their centers passes through the point of tangency, and its length is the sum (for external tangency) or difference (for internal tangency) of their radii.

11. Let $M_3(\mathbb{R})$ be the vector space of all 3×3 real matrices over $\mathbb{R}$ under usual matrix addition and scalar multiplication. Let $T_3(\mathbb{R})$ be the set of all 3×3 real upper triangular matrices. Which one of the following is TRUE?

- (A) The quotient space $M_3(\mathbb{R})/T_3(\mathbb{R})$ is isomorphic to the vector space of all 3×3 real skew symmetric matrices over $\mathbb{R}$ under usual matrix addition and scalar multiplication.
- (B) The quotient space $M_3(\mathbb{R})/T_3(\mathbb{R})$ is isomorphic to the vector space of all 3×3 real symmetric matrices over $\mathbb{R}$ under usual matrix addition and scalar multiplication.
- (C) The quotient space $M_3(\mathbb{R})/T_3(\mathbb{R})$ is isomorphic to the vector space of all 3×3 real lower triangular matrices over $\mathbb{R}$ under usual matrix addition and scalar multiplication.
- (D) The quotient space $M_3(\mathbb{R})/T_3(\mathbb{R})$ is isomorphic to the vector space of all 3×3 real matrices with trace zero over $\mathbb{R}$ under usual matrix addition and scalar multiplication.

Correct Answer: (A) The quotient space $M_3(\mathbb{R})/T_3(\mathbb{R})$ is isomorphic to the vector space of all 3×3 real skew symmetric matrices over $\mathbb{R}$ under usual matrix addition and scalar multiplication.

Solution:

Step 1: Understanding the Question:

The question asks to identify which vector space is isomorphic to the quotient space $M_3(\mathbb{R})/T_3(\mathbb{R})$. Two finite-dimensional vector spaces are isomorphic if and only if they have the same dimension.

Step 2: Key Formula or Approach:

The dimension of a quotient space V/W is given by $\dim(V/W) = \dim(V) - \dim(W)$. We need to calculate the dimension of the quotient space and then compare it with the dimensions of the vector spaces given in the options.

Step 3: Detailed Explanation:

1. Dimension of $M_3(\mathbb{R})$:

The vector space $M_3(\mathbb{R})$ consists of all 3×3 matrices. It has $3 \times 3 = 9$ entries, so its dimension is 9.

$$\dim(M_3(\mathbb{R})) = 9$$

2. Dimension of $T_3(\mathbb{R})$:

The vector space $T_3(\mathbb{R})$ consists of all 3×3 upper triangular matrices. The entries below the main diagonal are zero. The number of non-zero entries can be chosen freely, which are on and above the main diagonal. For a 3×3 matrix, this is $3 + 2 + 1 = 6$ entries.

$$\dim(T_3(\mathbb{R})) = 6$$

3. Dimension of the Quotient Space:

$$\dim(M_3(\mathbb{R})/T_3(\mathbb{R})) = \dim(M_3(\mathbb{R})) - \dim(T_3(\mathbb{R})) = 9 - 6 = 3$$

4. Dimensions of the spaces in the options:

(A) **Skew-symmetric matrices:** A matrix A is skew-symmetric if $A^T = -A$. For a 3×3 matrix, this implies the diagonal elements are zero, and $a_{ij} = -a_{ji}$. We only need to choose the entries above the main diagonal (a_{12}, a_{13}, a_{23}). The number of independent entries is $\frac{n(n-1)}{2} = \frac{3(2)}{2} = 3$. So, the dimension is 3.

(B) **Symmetric matrices:** A matrix A is symmetric if $A^T = A$. We can choose the entries on and above the main diagonal freely. The number of independent entries is $\frac{n(n+1)}{2} = \frac{3(4)}{2} = 6$. The dimension is 6.

(C) **Lower triangular matrices:** Similar to upper triangular matrices, the dimension is 6.

(D) **Matrices with trace zero:** The trace is the sum of the diagonal elements, $\text{tr}(A) = a_{11} + a_{22} + a_{33} = 0$. This is one linear constraint on the 9 entries of the matrix. So the dimension is $9 - 1 = 8$.

Step 4: Final Answer:

The dimension of the quotient space is 3. Among the given options, only the vector space of 3×3 real skew-symmetric matrices has a dimension of 3. Therefore, the two spaces are isomorphic.

💡 Quick Tip

To quickly find the dimension of a space of matrices with certain properties (symmetric, skew-symmetric, triangular, etc.), just count the number of entries you can choose freely. For a quotient space V/W , remember the dimension formula: $\dim(V) - \dim(W)$.

12. Let I be the integral defined as follows:

$$I = \int_0^1 \int_{x^2}^{x^2+1} dy dx$$

I am assuming this from the provided solution, as the question in the PDF seems to be different and likely contains a typo. If the question was to change the order of integration for

$$I = \int_0^1 \int_0^{\sqrt{y}} f(x, y) dx dy + \int_1^2 \int_{\sqrt{y-1}}^1 f(x, y) dx dy$$

then the correct changed order would be: Which one of the following is the correct expression for I?

- (A) $\int_0^{1/2} \int_{-x}^x dy dx + \int_{1/2}^1 \int_{-x^2+1}^{x^2+1} dy dx$
- (B) $\int_0^1 \int_{x^2}^{x^2+1} dy dx$
- (C) $\int_0^{1/2} \int_x^{x^2+1} dy dx + \int_{1/2}^1 \int_{\sqrt{x}}^{\sqrt{x+1}} dy dx$
- (D) $\int_0^1 \int_{x^2}^{x^2-1} dy dx$

Correct Answer: (B) $\int_0^1 \int_{x^2}^{x^2+1} dy dx$

Solution:

Step 1: Understanding the Question:

We are asked to change the order of integration. We will analyze the region of integration described by the integral in option (B) and then express it in the $dx dy$ order.

Step 2: Key Approach: Sketching the Region of Integration

The integral given in option (B) is $I = \int_0^1 \int_{x^2}^{x^2+1} dy dx$.

The region of integration R is described by the inequalities:

$$0 \leq x \leq 1$$

$$x^2 \leq y \leq x^2 + 1$$

Step 3: Detailed Explanation:

Let's sketch the boundaries of the region R :

- $x = 0$ (the y-axis)
- $x = 1$ (a vertical line)
- $y = x^2$ (a parabola opening upwards, passing through (0,0) and (1,1))
- $y = x^2 + 1$ (the same parabola shifted up by 1, passing through (0,1) and (1,2))

The region R is the area between the two parabolas from $x = 0$ to $x = 1$. To change the order of integration to $dx dy$, we need to express the bounds of x in terms of y . Looking at the region, we see that the shape of the boundaries for x changes as y crosses the line $y = 1$. So we must split the integral into two parts: one for $0 \leq y \leq 1$ and one for $1 \leq y \leq 2$.

Part 1: Region for $0 \leq y \leq 1$

In this range, the region is bounded on the left by $x = 0$ and on the right by the curve $y = x^2$, which gives $x = \sqrt{y}$.

So, for $0 \leq y \leq 1$, the bounds for x are $0 \leq x \leq \sqrt{y}$.

Part 2: Region for $1 \leq y \leq 2$

In this range, the region is bounded on the left by the curve $y = x^2 + 1$, which gives $x = \sqrt{y-1}$, and on the right by the line $x = 1$.

So, for $1 \leq y \leq 2$, the bounds for x are $\sqrt{y-1} \leq x \leq 1$.

Combining these two parts, the integral I with the order of integration changed is:

$$I = \int_0^1 \int_0^{\sqrt{y}} dx dy + \int_1^2 \int_{\sqrt{y-1}}^1 dx dy$$

This shows that the integral in option (B) is the correct single-integral representation in $dydx$ order for a region that would be split into two integrals in $dx dy$ order. The question likely asked for this conversion.

Step 4: Final Answer:

The correct expression representing the described region in the $dydx$ order is $\int_0^1 \int_{x^2}^{x^2+1} dy dx$.

💡 Quick Tip

When changing the order of integration, always sketch the region first. If the bounding curves change definition across the region, you will need to split the integral into multiple parts. Look for "corners" or points where the left/right or top/bottom boundary functions change.

13. Let C denote the Cantor set and $f : [0, 1] \rightarrow \mathbb{R}$ be defined as follows:

$$f(x) = \begin{cases} x^{2026} & \text{for } x \in C \\ \cos(\pi x) & \text{for } x \in [0, \frac{1}{2}] \setminus C \\ \sin(\pi x) & \text{for } x \in (\frac{1}{2}, 1] \setminus C \end{cases}$$

The value of the Lebesgue integral of $f(x)$ over the interval $[0,1]$ is equal to

- (A) $\frac{2}{\pi}$
- (B) $\frac{1}{\pi}$
- (C) $\frac{3}{\pi}$
- (D) 0

Correct Answer: (A) $\frac{2}{\pi}$

Solution:

Step 1: Understanding the Question:

We need to calculate the Lebesgue integral of a function $f(x)$ defined piecewise over the interval

$[0,1]$. The definition of the function depends on the Cantor set C .

Step 2: Key Formula or Approach:

The Lebesgue integral has a key property: the integral of any function over a set of measure zero is zero. The standard Cantor set C is an uncountable set with Lebesgue measure zero, i.e., $m(C) = 0$.

The integral over a union of disjoint sets is the sum of the integrals over those sets. We will split the domain $[0,1]$ into three disjoint sets: C , $[0, 1/2] \setminus C$, and $(\frac{1}{2}, 1] \setminus C$.

Step 3: Detailed Explanation:

The Lebesgue integral of f over $[0,1]$ can be written as:

$$\int_{[0,1]} f \, dm = \int_C f \, dm + \int_{[0,1/2] \setminus C} f \, dm + \int_{(\frac{1}{2}, 1] \setminus C} f \, dm$$

1. Integral over the Cantor set C :

The Lebesgue measure of the Cantor set is $m(C) = 0$. Since the function $f(x) = x^{2026}$ is bounded on C , the integral over this set is zero.

$$\int_C f \, dm = \int_C x^{2026} \, dm = 0$$

2. Integral over $[0, 1/2] \setminus C$:

On this set, $f(x) = \cos(\pi x)$. Since the set C has measure zero, removing it from the interval $[0, 1/2]$ does not change the value of the integral. Also, for a continuous function like $\cos(\pi x)$, the Lebesgue integral is the same as the Riemann integral.

$$\begin{aligned} \int_{[0,1/2] \setminus C} \cos(\pi x) \, dm &= \int_0^{1/2} \cos(\pi x) \, dx \\ &= \left[\frac{\sin(\pi x)}{\pi} \right]_0^{1/2} = \frac{\sin(\pi/2)}{\pi} - \frac{\sin(0)}{\pi} = \frac{1}{\pi} - 0 = \frac{1}{\pi} \end{aligned}$$

3. Integral over $(\frac{1}{2}, 1] \setminus C$:

On this set, $f(x) = \sin(\pi x)$. Similar to the previous case, we can compute the Riemann integral over the interval $(\frac{1}{2}, 1]$.

$$\begin{aligned} \int_{(\frac{1}{2}, 1] \setminus C} \sin(\pi x) \, dm &= \int_{1/2}^1 \sin(\pi x) \, dx \\ &= \left[-\frac{\cos(\pi x)}{\pi} \right]_{1/2}^1 = \left(-\frac{\cos(\pi)}{\pi} \right) - \left(-\frac{\cos(\pi/2)}{\pi} \right) = \left(-\frac{-1}{\pi} \right) - \left(-\frac{0}{\pi} \right) = \frac{1}{\pi} \end{aligned}$$

Total Integral:

The total value is the sum of the three parts:

$$\int_{[0,1]} f \, dm = 0 + \frac{1}{\pi} + \frac{1}{\pi} = \frac{2}{\pi}$$

Step 4: Final Answer:

The value of the Lebesgue integral is $\frac{2}{\pi}$.

 Quick Tip

Remembering that the Cantor set has measure zero is crucial for many problems in real analysis and measure theory. This property often simplifies calculations by allowing you to ignore the part of the integral over the Cantor set itself.

14. Let G be a group of order 595. Which one of the following is TRUE?

- (A) G cannot have a proper normal subgroup.
- (B) G must have a proper normal subgroup.
- (C) G cannot have an element of order 17.
- (D) The number of Sylow 5-subgroups is 17.

Correct Answer: (B) G must have a proper normal subgroup.

Solution:

Step 1: Understanding the Question:

We are asked to determine a property of a group G with order $|G| = 595$. The properties relate to the existence of proper normal subgroups and elements of a specific order.

Step 2: Key Formula or Approach:

We will use Sylow's Theorems to analyze the structure of the group. The first step is to find the prime factorization of the order of the group.

Sylow's Third Theorem states that if $|G| = p^k m$ where p does not divide m , then the number of Sylow p -subgroups, n_p , satisfies:

1. $n_p \equiv 1 \pmod{p}$
2. n_p divides m .

A key consequence is that if $n_p = 1$, the unique Sylow p -subgroup is a normal subgroup.

Step 3: Detailed Explanation:

First, find the prime factorization of 595:

$$595 = 5 \times 119 = 5 \times 7 \times 17$$

The order of the group is $|G| = 5 \cdot 7 \cdot 17$. Let's analyze the number of Sylow p -subgroups for each prime factor.

- Sylow 5-subgroups (n_5):

$n_5 \equiv 1 \pmod{5}$ and n_5 must divide $7 \times 17 = 119$.

The divisors of 119 are 1, 7, 17, 119.

Checking the congruence condition:

- $1 \equiv 1 \pmod{5}$ (True)
- $7 \equiv 2 \pmod{5}$ (False)
- $17 \equiv 2 \pmod{5}$ (False)

$119 \equiv 4 \pmod{5}$ (False)

The only possibility is $n_5 = 1$.

- **Sylow 7-subgroups (n_7):**

$n_7 \equiv 1 \pmod{7}$ and n_7 must divide $5 \times 17 = 85$.

The divisors of 85 are 1, 5, 17, 85.

Checking the congruence condition:

$1 \equiv 1 \pmod{7}$ (True)

$5 \equiv 5 \pmod{7}$ (False)

$17 = 2 \times 7 + 3 \implies 17 \equiv 3 \pmod{7}$ (False)

$85 = 12 \times 7 + 1 \implies 85 \equiv 1 \pmod{7}$ (True)

So, n_7 could be 1 or 85. - **Sylow 17-subgroups (n_{17}):**

$n_{17} \equiv 1 \pmod{17}$ and n_{17} must divide $5 \times 7 = 35$.

The divisors of 35 are 1, 5, 7, 35.

Checking the congruence condition:

$1 \equiv 1 \pmod{17}$ (True)

$5 \equiv 5 \pmod{17}$ (False)

$7 \equiv 7 \pmod{17}$ (False)

$35 = 2 \times 17 + 1 \implies 35 \equiv 1 \pmod{17}$ (True)

So, n_{17} could be 1 or 35. Since we found that $n_5 = 1$, there exists a unique Sylow 5-subgroup.

A unique Sylow p -subgroup is always a normal subgroup. This subgroup has order 5. Since $1 < 5 < 595$, this is a proper normal subgroup.

Now let's evaluate the options:

(A) G cannot have a proper normal subgroup. This is FALSE, as we found a normal subgroup of order 5.

(B) G must have a proper normal subgroup. This is TRUE.

(C) G cannot have an element of order 17. This is FALSE by Cauchy's Theorem, which guarantees an element of order 17 since 17 is a prime dividing the order of G .

(D) The number of Sylow 5-subgroups is 17. This is FALSE, we calculated $n_5 = 1$.

Step 4: Final Answer:

The group G must have a proper normal subgroup (the Sylow 5-subgroup).

💡 Quick Tip

When analyzing a group of a given order, always start with its prime factorization. Then, apply Sylow's Theorems to find the possible number of Sylow p -subgroups. If $n_p = 1$ for any prime p , you've found a normal subgroup.

15. Let $u(x, t)$ be the solution of the initial value problem for the heat equation on the real line:

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, \quad -\infty < x < \infty, \quad t > 0, \quad k \in \mathbb{R}$$

with the initial condition

$$u(x, 0) = e^{-a|x|}, \quad a > 0.$$

If $\hat{u}(\omega, t) = \int_{-\infty}^{\infty} u(x, t)e^{i\omega x} dx$ is the Fourier transform of $u(x, t)$ with respect to x , then $\hat{u}(\omega, t)$ is equal to

- (A) $\frac{2a}{a^2 + \omega^2} e^{-k\omega^2 t}$
- (B) $\frac{a}{a^2 + \omega^2} e^{-k\omega^2 t}$
- (C) $\frac{2a}{a^2 + \omega^2} e^{-ka^2 t}$
- (D) $\frac{1}{\sqrt{4\pi kt}} e^{\frac{-\omega^2}{4kt}}$

Correct Answer: (A) $\frac{2a}{a^2 + \omega^2} e^{-k\omega^2 t}$

Solution:

Step 1: Understanding the Question:

We need to solve the heat equation with a given initial condition using the Fourier transform method. The goal is to find the Fourier transform of the solution, $\hat{u}(\omega, t)$.

Step 2: Key Formula or Approach:

1. Apply the Fourier transform to the PDE to convert it into an ODE in time t .
 2. Solve the resulting ODE.
 3. Use the Fourier transform of the initial condition to determine the constant of integration.
- The required Fourier transform properties are (using the definition $\mathcal{F}\{f(x)\} = \int_{-\infty}^{\infty} f(x)e^{i\omega x} dx$):
- $\mathcal{F}\left\{\frac{\partial u}{\partial t}\right\} = \frac{d\hat{u}}{dt}$
 - $\mathcal{F}\left\{\frac{\partial^2 u}{\partial x^2}\right\} = (-i\omega)^2 \hat{u}(\omega, t) = -\omega^2 \hat{u}(\omega, t)$

Step 3: Detailed Explanation:

1. Transform the PDE:

Applying the Fourier transform to the heat equation $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$, we get:

$$\frac{d\hat{u}(\omega, t)}{dt} = k(-\omega^2 \hat{u}(\omega, t))$$

$$\frac{d\hat{u}}{dt} = -k\omega^2 \hat{u}$$

This is a first-order linear ordinary differential equation for $\hat{u}$ as a function of t , with ω as a parameter.

2. Solve the ODE:

The general solution is:

$$\hat{u}(\omega, t) = C(\omega)e^{-k\omega^2 t}$$

where $C(\omega)$ is the constant of integration, which depends on ω .

3. Apply the Initial Condition:

To find $C(\omega)$, we set $t = 0$:

$$\hat{u}(\omega, 0) = C(\omega)e^0 = C(\omega)$$

We find $\hat{u}(\omega, 0)$ by taking the Fourier transform of the initial condition $u(x, 0) = e^{-a|x|}$.

$$\hat{u}(\omega, 0) = \mathcal{F}\{u(x, 0)\} = \int_{-\infty}^{\infty} e^{-a|x|} e^{i\omega x} dx$$

We split the integral:

$$\begin{aligned} &= \int_{-\infty}^0 e^{ax} e^{i\omega x} dx + \int_0^{\infty} e^{-ax} e^{i\omega x} dx \\ &= \int_{-\infty}^0 e^{(a+i\omega)x} dx + \int_0^{\infty} e^{(-a+i\omega)x} dx \\ &= \left[\frac{e^{(a+i\omega)x}}{a+i\omega} \right]_{-\infty}^0 + \left[\frac{e^{(-a+i\omega)x}}{-a+i\omega} \right]_0^{\infty} \end{aligned}$$

Assuming $a > 0$, the terms at $\pm\infty$ go to zero.

$$\begin{aligned} &= \left(\frac{1}{a+i\omega} - 0 \right) + \left(0 - \frac{1}{-a+i\omega} \right) \\ &= \frac{1}{a+i\omega} + \frac{1}{a-i\omega} \\ &= \frac{(a-i\omega) + (a+i\omega)}{(a+i\omega)(a-i\omega)} = \frac{2a}{a^2 - (i\omega)^2} = \frac{2a}{a^2 + \omega^2} \end{aligned}$$

So, $C(\omega) = \hat{u}(\omega, 0) = \frac{2a}{a^2 + \omega^2}$.

4. Final Solution for $\hat{u}(\omega, t)$:

Substitute $C(\omega)$ back into the general solution:

$$\hat{u}(\omega, t) = \frac{2a}{a^2 + \omega^2} e^{-k\omega^2 t}$$

Step 4: Final Answer:

The Fourier transform of the solution is $\frac{2a}{a^2 + \omega^2} e^{-k\omega^2 t}$.

💡 Quick Tip

The Fourier transform is a powerful tool for solving linear PDEs on an infinite domain. It converts differentiation with respect to the spatial variable x into multiplication by $i\omega$ (or $-i\omega$), transforming the PDE into a simpler ODE. Memorize the Fourier transforms of common functions like the Gaussian e^{-ax^2} and the exponential decay $e^{-a|x|}$.

16. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined as $f(x, y) = (e^x \cos y, e^x \sin y)$. Which one of the following is TRUE?

- (A) f is one-to-one.
- (B) The Jacobian of f is negative.

- (C) f is locally invertible.
 (D) f is invertible on $\mathbb{R}^2$.

Correct Answer: (C) f is locally invertible.

Solution:

Step 1: Understanding the Question:

We need to analyze the properties (injectivity, invertibility, local invertibility, and Jacobian sign) of the given function $f(x, y)$. This function can be recognized as the mapping from Cartesian coordinates (x, y) to polar coordinates with radius $r = e^x$ and angle $\theta = y$, or equivalently, the complex exponential function $z \mapsto e^z$ where $z = x + iy$.

Step 2: Key Formula or Approach:

- To check if f is one-to-one, we see if $f(x_1, y_1) = f(x_2, y_2)$ implies $(x_1, y_1) = (x_2, y_2)$.
- Invertibility on $\mathbb{R}^2$ requires the function to be both one-to-one and onto.
- Local invertibility is determined by the Inverse Function Theorem, which states that if the Jacobian determinant is non-zero at a point, the function is invertible in a neighborhood of that point.
- The Jacobian determinant is calculated as $\det \left(\frac{\partial(u,v)}{\partial(x,y)} \right)$, where $u = e^x \cos y$ and $v = e^x \sin y$.

Step 3: Detailed Explanation:

(A) Is f one-to-one?

Let's test two different points. Consider $(x_1, y_1) = (0, 0)$ and $(x_2, y_2) = (0, 2\pi)$.

$$f(0, 0) = (e^0 \cos 0, e^0 \sin 0) = (1, 0).$$

$$f(0, 2\pi) = (e^0 \cos 2\pi, e^0 \sin 2\pi) = (1, 0).$$

Since $f(0, 0) = f(0, 2\pi)$ but $(0, 0) \neq (0, 2\pi)$, the function is not one-to-one. It is periodic in y with period 2π . So, (A) is false.

(D) Is f invertible on $\mathbb{R}^2$?

A function must be one-to-one to be invertible. Since f is not one-to-one, it is not invertible on $\mathbb{R}^2$. So, (D) is false.

(B) The Jacobian of f :

Let $u(x, y) = e^x \cos y$ and $v(x, y) = e^x \sin y$. The Jacobian matrix is:

$$J_f = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix}$$

The Jacobian determinant is:

$$\begin{aligned} \det(J_f) &= (e^x \cos y)(e^x \cos y) - (-e^x \sin y)(e^x \sin y) \\ &= e^{2x} \cos^2 y + e^{2x} \sin^2 y \\ &= e^{2x} (\cos^2 y + \sin^2 y) = e^{2x} \end{aligned}$$

Since e^{2x} is always positive for any real x , the Jacobian is always positive. Therefore, the statement "The Jacobian of f is negative" is false. So, (B) is false.

(C) Is f locally invertible?

According to the Inverse Function Theorem, a function f is locally invertible at any point where its Jacobian determinant is non-zero.

We found that the Jacobian determinant is e^{2x} , which is never zero for any $(x, y) \in \mathbb{R}^2$. Therefore, f is locally invertible everywhere on $\mathbb{R}^2$. So, (C) is true.

Step 4: Final Answer:

The only true statement is that f is locally invertible.

💡 Quick Tip

The function $f(x, y) = (e^x \cos y, e^x \sin y)$ is the real-variable representation of the complex exponential function $f(z) = e^z$. Remembering its properties, like being periodic (hence not one-to-one globally) but having a non-zero derivative ($e^z \neq 0$, which corresponds to a non-zero Jacobian), can help you answer these questions quickly.

17. Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be defined by $f(z) = |z|^2 - 5\bar{z} + 2$. Which one of the following is TRUE?

- (A) f is differentiable at $z = 5i$.
- (B) f is differentiable at $z = 5$.
- (C) f is differentiable at $z = -5i$.
- (D) f is differentiable at $z = -5$.

Correct Answer: (B) f is differentiable at $z = 5$.

Solution:

Step 1: Understanding the Question:

We are given a complex function $f(z)$ that involves $|z|^2$ and $\bar{z}$. We need to find the point(s) where f is differentiable. Such functions are generally not analytic (differentiable everywhere), but they might be differentiable at specific points.

Step 2: Key Formula or Approach:

A complex function $f(z)$ is differentiable at a point z_0 if and only if the Cauchy-Riemann (CR) equations are satisfied at that point. Since the function involves $\bar{z}$, a convenient way to apply the CR equations is to use the Wirtinger derivatives. The necessary condition for differentiability is $\frac{\partial f}{\partial \bar{z}} = 0$.

We use the relations: $z = x + iy$, $\bar{z} = x - iy$, and $|z|^2 = z\bar{z}$. Also, $x = \frac{z+\bar{z}}{2}$ and $y = \frac{z-\bar{z}}{2i}$.

Step 3: Detailed Explanation:

The given function is $f(z) = |z|^2 - 5\bar{z} + 2$.

Substitute $|z|^2 = z\bar{z}$:

$$f(z) = z\bar{z} - 5\bar{z} + 2$$

To find where f is differentiable, we compute the partial derivative of f with respect to $\bar{z}$, treating z and $\bar{z}$ as independent variables.

$$\begin{aligned}\frac{\partial f}{\partial \bar{z}} &= \frac{\partial}{\partial \bar{z}}(z\bar{z} - 5\bar{z} + 2) \\ \frac{\partial f}{\partial \bar{z}} &= z \cdot \frac{\partial \bar{z}}{\partial \bar{z}} - 5 \cdot \frac{\partial \bar{z}}{\partial \bar{z}} + \frac{\partial 2}{\partial \bar{z}} \\ \frac{\partial f}{\partial \bar{z}} &= z - 5 + 0 = z - 5\end{aligned}$$

The function f is differentiable only at the points where $\frac{\partial f}{\partial \bar{z}} = 0$. Setting the derivative to zero:

$$z - 5 = 0$$

$$z = 5$$

So, the function is differentiable only at the point $z = 5$.

Let's check the options: (A) $z = 5i$. Here $z - 5 = 5i - 5 \neq 0$. Not differentiable.

(B) $z = 5$. Here $z - 5 = 5 - 5 = 0$. Differentiable.

(C) $z = -5i$. Here $z - 5 = -5i - 5 \neq 0$. Not differentiable.

(D) $z = -5$. Here $z - 5 = -5 - 5 = -10 \neq 0$. Not differentiable.

Step 4: Final Answer:

The function $f(z)$ is differentiable at $z = 5$.

💡 Quick Tip

For complex functions involving $\bar{z}$ or $|z|^2$, the Wirtinger derivative $\frac{\partial f}{\partial \bar{z}}$ is a very efficient tool. Remember that a function is analytic if and only if $\frac{\partial f}{\partial \bar{z}} = 0$ in an open set, and it is differentiable at a point z_0 if $\frac{\partial f}{\partial \bar{z}} = 0$ at z_0 .

18. Let $D = \{z \in \mathbb{C} \mid |z| < 1\}$ denote the unit disc in the complex plane $\mathbb{C}$. Let $f : D \rightarrow D$ be an analytic function which satisfies $f(0) = 0$. Then which one of the following is a possible value of $f'(0)$?

- (A) $\frac{5}{2}i$
- (B) $\frac{i}{10}$
- (C) $\frac{3}{2}$
- (D) $-\frac{5}{2}i$

Correct Answer: (B) $\frac{i}{10}$

Solution:**Step 1: Understanding the Question:**

We have an analytic function f that maps the unit disc D into itself, i.e., $|f(z)| \leq 1$ for $|z| < 1$. Additionally, we are given that the function maps the origin to itself, $f(0) = 0$. We need to determine a possible value for the derivative at the origin, $f'(0)$.

Step 2: Key Formula or Approach:

This is a direct application of the Schwarz Lemma.

Schwarz Lemma: Let $f : D \rightarrow \mathbb{C}$ be an analytic function such that:

1. $|f(z)| \leq 1$ for all $z \in D$ (i.e., $f(D) \subseteq D$).
2. $f(0) = 0$.

Then, $|f(z)| \leq |z|$ for all $z \in D$, and $|f'(0)| \leq 1$.

Step 3: Detailed Explanation:

The conditions of the problem perfectly match the hypotheses of the Schwarz Lemma.

1. f is analytic on $D = \{z : |z| < 1\}$.
2. The range of f is D , so $|f(z)| < 1$ for all $z \in D$.
3. $f(0) = 0$.

According to the Schwarz Lemma, the magnitude of the derivative at the origin must be less than or equal to 1.

$$|f'(0)| \leq 1$$

We need to check the magnitudes of the values given in the options.

- (A) $f'(0) = \frac{5}{2}i$. The magnitude is $\left|\frac{5}{2}i\right| = \frac{5}{2} = 2.5$. Since $2.5 > 1$, this is not a possible value.
- (B) $f'(0) = \frac{i}{10}$. The magnitude is $\left|\frac{i}{10}\right| = \frac{1}{10} = 0.1$. Since $0.1 \leq 1$, this is a possible value.
- (C) $f'(0) = \frac{3}{2}$. The magnitude is $\left|\frac{3}{2}\right| = \frac{3}{2} = 1.5$. Since $1.5 > 1$, this is not a possible value.
- (D) $f'(0) = -\frac{5}{2}i$. The magnitude is $\left|-\frac{5}{2}i\right| = \frac{5}{2} = 2.5$. Since $2.5 > 1$, this is not a possible value.

Step 4: Final Answer:

The only value whose magnitude is less than or equal to 1 is $\frac{i}{10}$. Therefore, this is the only possible value for $f'(0)$.

💡 Quick Tip

Whenever you see a problem involving an analytic function mapping the unit disk to itself with $f(0) = 0$, immediately think of the Schwarz Lemma. The key results are $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$.

19. Let X be the space of all continuously differentiable real valued functions on $[0,1]$. Define the following norms on X :

$$p_1(x) = \sup\{|x(t)| : t \in [0, 1]\}$$

$$p_2(x) = \sup\left\{\left|\frac{d}{dt}x(t)\right| : t \in [0, 1]\right\}$$

$$\text{and } p_3(x) = p_1(x) + p_2(x).$$

Which one of the following is TRUE?

- (A) (X, p_1) is a Banach space.
- (B) (X, p_2) is a Banach space.
- (C) (X, p_3) is not a Banach space.
- (D) (X, p_3) is a Banach space.

Correct Answer: (D) (X, p_3) is a Banach space.

Solution:

Step 1: Understanding the Question:

We are given a vector space $X = C^1[0, 1]$ (continuously differentiable functions on $[0,1]$) and three different norms defined on it. We need to determine which of the resulting normed spaces are Banach spaces. A Banach space is a complete normed vector space.

Step 2: Key Formula or Approach:

To check if a normed space is a Banach space, we must verify if every Cauchy sequence in the space converges to a limit that is also in the space.

Step 3: Detailed Explanation:

Let's analyze each option.

(A) (X, p_1) :

$X = C^1[0, 1]$ with the sup-norm $p_1(x) = \|x\|_\infty$. We need to check for completeness. Consider a sequence of functions (x_n) in $C^1[0, 1]$ that is Cauchy with respect to p_1 . The space of all continuous functions $C[0, 1]$ with the sup-norm is a Banach space. Therefore, the sequence (x_n) converges uniformly to a continuous function $x \in C[0, 1]$. The question is whether this limit function x must be in $C^1[0, 1]$, i.e., must be continuously differentiable. The answer is no. For example, a sequence of smooth polynomials can converge uniformly to a non-differentiable function like $|t - 1/2|$. Thus, (X, p_1) is not complete. So, (A) is false.

(B) (X, p_2) :

$X = C^1[0, 1]$ with the norm $p_2(x) = \sup |x'(t)|$. First, we need to check if p_2 is actually a norm. A norm must satisfy $p_2(x) = 0 \iff x = 0$. If we take a constant function $x(t) = c \neq 0$, then $x'(t) = 0$, so $p_2(x) = 0$, but x is not the zero function. Hence, p_2 is not a norm on X . It is a semi-norm. So, (B) is false. (Even if we consider the quotient space where constants are identified with zero, the space is not complete).

(D) (X, p_3) :

Let $X = C^1[0, 1]$ with the norm $p_3(x) = \sup |x(t)| + \sup |x'(t)|$. Let (x_n) be a Cauchy sequence in (X, p_3) . This means that for any $\epsilon > 0$, there exists N such that for $m, n > N$,

$$p_3(x_n - x_m) = \sup |x_n(t) - x_m(t)| + \sup |x'_n(t) - x'_m(t)| < \epsilon$$

This implies that both (x_n) and (x'_n) are Cauchy sequences under the sup-norm.

1. Since (x_n) is Cauchy in the sup-norm, it converges uniformly to a continuous function $x \in C[0, 1]$.

2. Since (x'_n) is Cauchy in the sup-norm, it converges uniformly to a continuous function, let's call it $y \in C[0, 1]$.

Now we need to check if $x \in C^1[0, 1]$ and if $x' = y$. By a standard theorem in analysis (interchange of limit and derivative), if a sequence of differentiable functions (x_n) converges at some point (e.g., $x_n(0)$ converges), and the sequence of derivatives (x'_n) converges uniformly to y , then the original sequence (x_n) converges uniformly to a differentiable function x and $x' = y$.

In our case, uniform convergence of (x_n) implies pointwise convergence. So, the conditions are met. The limit function x is in $C^1[0, 1]$ (since its derivative y is continuous). The convergence $x_n \rightarrow x$ and $x'_n \rightarrow x'$ in the sup-norm implies $p_3(x_n - x) \rightarrow 0$.

Therefore, the space (X, p_3) is complete. It is a Banach space. So, (D) is true.

(C) (X, p_3) is not a Banach space.

Since we have shown (D) is true, (C) must be false.

Step 4: Final Answer:

(X, p_3) is a complete normed space, hence a Banach space.

 Quick Tip

To test if a function space is a Banach space, construct a Cauchy sequence and see if its limit retains the defining property of the space (e.g., differentiability, continuity). The space $C^k[a, b]$ of k -times continuously differentiable functions is a Banach space under the norm $\|f\|_{C^k} = \sum_{i=0}^k \sup |f^{(i)}(x)|$, which is equivalent to the norm p_3 for $k = 1$.

20. Let X be a set with at least two elements. Let d_1, d_2 and d_3 be metrics on X . Which one of the following is NOT a metric on X ?

- (A) $d(x, y) := \min(d_1(x, y), 3)$ for all $x, y \in X$.
- (B) $d(x, y) := \max(d_2(x, y), 3)$ for all $x, y \in X$.
- (C) $d(x, y) := \frac{10d_3(x, y)}{1+d_3(x, y)}$ for all $x, y \in X$.
- (D) $d(x, y) := \frac{1}{3}(d_1(x, y) + d_2(x, y) + d_3(x, y))$ for all $x, y \in X$.

Correct Answer: (B) $d(x, y) := \max(d_2(x, y), 3)$ for all $x, y \in X$.

Solution:

Step 1: Understanding the Question:

We are given four functions defined using known metrics d_1, d_2, d_3 . We need to identify which of these new functions fails to satisfy the properties of a metric.

Step 2: Key Formula or Approach:

A function $d : X \times X \rightarrow \mathbb{R}$ is a metric if for all $x, y, z \in X$, it satisfies:

1. **Non-negativity:** $d(x, y) \geq 0$.
2. **Identity of indiscernibles:** $d(x, y) = 0 \iff x = y$.
3. **Symmetry:** $d(x, y) = d(y, x)$.
4. **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$.

We will check these properties for each option.

Step 3: Detailed Explanation:

(A) $d(x, y) = \min(d_1(x, y), 3)$: This is a standard way to create a bounded metric from an unbounded one. It can be shown to satisfy all metric properties. This is a metric.

(B) $d(x, y) = \max(d_2(x, y), 3)$: Let's check the properties.

1. Non-negativity: Since $d_2(x, y) \geq 0$, $\max(d_2(x, y), 3)$ is always ≥ 3 . So, $d(x, y) \geq 0$. This holds.

2. Identity of indiscernibles: We need $d(x, y) = 0 \iff x = y$. Let's calculate $d(x, x)$. Since d_2 is a metric, $d_2(x, x) = 0$. Then $d(x, x) = \max(d_2(x, x), 3) = \max(0, 3) = 3$.

We have $d(x, x) = 3 \neq 0$. This violates the condition that $d(x, x)$ must be 0. Therefore, this function is not a metric.

(C) $d(x, y) = \frac{10d_3(x, y)}{1+d_3(x, y)}$: This is of the form $f(t) = \frac{10t}{1+t}$. The function $f(t)$ is non-decreasing and concave for $t \geq 0$. This transformation preserves the metric properties. Specifically, it can be shown that if d_3 is a metric, then d is also a metric. This is a metric.

(D) $d(x, y) = \frac{1}{3}(d_1(x, y) + d_2(x, y) + d_3(x, y))$: This is a positive linear combination of metrics. A sum (or positive linear combination) of metrics is always a metric.

1. Non-negativity: Sum of non-negatives is non-negative.
2. Identity: $d(x, y) = 0 \iff d_1, d_2, d_3$ are all 0, which happens $\iff x = y$.
3. Symmetry: Follows from symmetry of d_1, d_2, d_3 .
4. Triangle inequality: Follows from adding the triangle inequalities for d_1, d_2, d_3 .

$$d_1(x, z) \leq d_1(x, y) + d_1(y, z)$$

$$d_2(x, z) \leq d_2(x, y) + d_2(y, z)$$

$$d_3(x, z) \leq d_3(x, y) + d_3(y, z)$$

Summing and dividing by 3 gives $d(x, z) \leq d(x, y) + d(y, z)$. This is a metric.

Step 4: Final Answer:

The function in option (B) is not a metric because it violates the identity of indiscernibles property, specifically, $d(x, x) \neq 0$.

 Quick Tip

When testing for a metric, the "identity of indiscernibles" ($d(x, y) = 0 \iff x = y$) is often the easiest to check and the most likely to fail for constructed functions. Always check if $d(x, x) = 0$ holds.

21. Let P be a 3×3 real symmetric positive definite matrix. Which of the following statements is/are TRUE?

- (A) $Q^T P Q$ is positive definite for all nonzero $Q \in \mathbb{R}^{3 \times 3}$.
- (B) $Q^T P Q$ is positive semidefinite for all $Q \in \mathbb{R}^{3 \times 3}$.
- (C) $Q^T P Q$ is positive definite if $Q \in \mathbb{R}^{3 \times 3}$ is nonsingular.
- (D) $Q^T P Q$ is not positive semidefinite for some $Q \in \mathbb{R}^{3 \times 3}$.

Correct Answer: (B) $Q^T P Q$ is positive semidefinite for all $Q \in \mathbb{R}^{3 \times 3}$., (C) $Q^T P Q$ is positive definite if $Q \in \mathbb{R}^{3 \times 3}$ is nonsingular.

Solution:

Step 1: Understanding the Question:

We are given a symmetric positive definite matrix P and are asked about the properties of the matrix $A = Q^T P Q$ for a matrix Q . A matrix M is positive definite if $x^T M x > 0$ for all nonzero vectors x . It is positive semidefinite if $x^T M x \geq 0$ for all nonzero vectors x .

Step 2: Key Formula or Approach:

We will use the definition of positive definite and positive semidefinite matrices. Since P is positive definite, we know that for any nonzero vector $y \in \mathbb{R}^3$, we have $y^T P y > 0$. We need to check the sign of $x^T A x$ for a nonzero vector x , where $A = Q^T P Q$.

$$x^T A x = x^T (Q^T P Q) x = (x^T Q^T) P (Q x) = (Q x)^T P (Q x)$$

Let $y = Q x$. Then $x^T A x = y^T P y$.

Step 3: Detailed Explanation:

Let's analyze the expression $y^T P y$ where $y = Q x$.

Since P is positive definite, we know $y^T P y \geq 0$ for any vector y . If $y \neq 0$, then $y^T P y > 0$.

Case 1: General $Q \in \mathbb{R}^{3 \times 3}$

Let x be any nonzero vector in $\mathbb{R}^3$. The vector $y = Q x$ can be zero even if x is nonzero. This happens if x is in the null space of Q , which is possible if Q is a singular matrix.

- If $y = Q x \neq 0$, then $x^T (Q^T P Q) x = y^T P y > 0$.

- If $y = Q x = 0$, then $x^T (Q^T P Q) x = y^T P y = 0$.

So, for any nonzero x , we have $x^T (Q^T P Q) x \geq 0$. This is the definition of a positive semidefinite matrix.

This means $Q^T P Q$ is always positive semidefinite for any Q . This makes statement (B) TRUE.

Statement (D) claims it's not positive semidefinite for some Q , which contradicts our finding. So (D) is FALSE.

Case 2: Nonsingular Q

If Q is nonsingular (invertible), then its null space contains only the zero vector. So, if x is a nonzero vector, then $y = Qx$ is also a nonzero vector.

Since $y = Qx \neq 0$, we have $y^T P y > 0$.

This means $x^T (Q^T P Q) x > 0$ for all nonzero x . This is the definition of a positive definite matrix.

So, if Q is nonsingular, $Q^T P Q$ is positive definite. This makes statement (C) TRUE.

Case 3: Nonzero singular Q

Let's reconsider statement (A), which claims $Q^T P Q$ is positive definite for all *nonzero* Q . We can choose a nonzero matrix Q that is singular. For example,

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

This matrix is nonzero. Let $x = (0, 0, 1)^T$. This is a nonzero vector. Then $y = Qx = (0, 0, 0)^T$. So, $x^T (Q^T P Q) x = y^T P y = 0$. Since we found a nonzero vector x for which the quadratic form is zero, the matrix $Q^T P Q$ is not positive definite. It is only positive semidefinite. Therefore, statement (A) is FALSE.

Step 4: Final Answer:

The true statements are (B) and (C).

💡 Quick Tip

The transformation $P \mapsto Q^T P Q$ is called a congruence transformation. Congruence transformations preserve symmetry. If Q is nonsingular, they also preserve definiteness (positive definite, negative definite, etc.), a result known as Sylvester's Law of Inertia. If Q is singular, the definiteness can be "weakened" (e.g., from positive definite to positive semidefinite).

22. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function defined as follows:

$$f(x, y) = \begin{cases} \frac{x\sqrt{x^2+y^2}}{|x|} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$$

Which of the following statements is/are TRUE?

- (A) $f(x, y)$ is continuous at $(0, 0)$.
- (B) For any $u \in \mathbb{R}^2$, the directional derivative of f at $(0, 0)$ in the direction of u exists.
- (C) $\frac{\partial f}{\partial x}(0, 0) = 1 = \frac{\partial f}{\partial y}(0, 0)$.

(D) $f(x, y)$ is differentiable at $(0,0)$.

Correct Answer: (A) $f(x, y)$ is continuous at $(0,0)$., (B) For any $u \in \mathbb{R}^2$, the directional derivative of f at $(0,0)$ in the direction of u exists.

Solution:

Step 1: Understanding the Question:

We are given a piecewise function $f(x, y)$ and need to analyze its continuity, directional derivatives, partial derivatives, and differentiability at the origin $(0,0)$.

Step 2: Simplify the Function Definition

For $x \neq 0$, the term $\frac{x}{|x|}$ is the sign function of x , denoted $\text{sgn}(x)$.

$$\text{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

So the function can be written as:

$$f(x, y) = \begin{cases} \sqrt{x^2 + y^2} & \text{for } x > 0 \\ -\sqrt{x^2 + y^2} & \text{for } x < 0 \\ 0 & \text{for } x = 0 \end{cases}$$

Step 3: Detailed Explanation:

(A) Continuity at $(0,0)$:

We need to check if $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = f(0, 0) = 0$.

Let's use polar coordinates: $x = r \cos \theta, y = r \sin \theta$.

Then $\sqrt{x^2 + y^2} = r$.

As $(x, y) \rightarrow (0, 0)$, we have $r \rightarrow 0$.

The magnitude of $f(x, y)$ for $x \neq 0$ is $|f(x, y)| = |\pm \sqrt{x^2 + y^2}| = \sqrt{x^2 + y^2} = r$.

As $r \rightarrow 0$, $|f(x, y)| \rightarrow 0$.

When $x = 0$, $f(0, y) = 0$. The limit is also 0 along this path. Since $|f(x, y)| \leq \sqrt{x^2 + y^2}$, and $\lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + y^2} = 0$, by the Squeeze Theorem, $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$.

Since the limit equals the function value $f(0, 0) = 0$, the function is continuous at $(0,0)$. So, (A) is TRUE.

(B) Directional Derivatives at $(0,0)$:

Let $u = (u_1, u_2)$ be a unit vector. The directional derivative is:

$$D_u f(0, 0) = \lim_{h \rightarrow 0} \frac{f(0 + hu_1, 0 + hu_2) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{f(hu_1, hu_2)}{h}$$

Case 1: $u_1 \neq 0$.

$$f(hu_1, hu_2) = \frac{hu_1}{|hu_1|} \sqrt{(hu_1)^2 + (hu_2)^2} = \frac{h}{|h|} \frac{u_1}{|u_1|} |h| \sqrt{u_1^2 + u_2^2} = h \frac{u_1}{|u_1|} \cdot 1$$

So, $D_u f(0, 0) = \lim_{h \rightarrow 0} \frac{h \cdot \text{sgn}(u_1)}{h} = \text{sgn}(u_1) = \frac{u_1}{|u_1|}$. The limit exists.

Case 2: $u_1 = 0$. Then $x = 0$. The unit vector must be $u = (0, 1)$ or $u = (0, -1)$. For $u = (0, 1)$, we approach along the y-axis. $f(0, h) = 0$.

$$D_u f(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0$$

. The limit exists.

Since the directional derivative exists for all unit vectors u , (B) is TRUE.

(C) Partial Derivatives at (0,0):

The partial derivative w.r.t x is the directional derivative in the direction $u = (1, 0)$. Here $u_1 = 1 > 0$.

$$\frac{\partial f}{\partial x}(0, 0) = D_{(1,0)} f(0, 0) = \text{sgn}(1) = 1.$$

The partial derivative w.r.t y is the directional derivative in the direction $u = (0, 1)$. Here $u_1 = 0$.

$$\frac{\partial f}{\partial y}(0, 0) = D_{(0,1)} f(0, 0) = 0.$$

The statement says $\frac{\partial f}{\partial x}(0, 0) = 1 = \frac{\partial f}{\partial y}(0, 0)$. This is false because $\frac{\partial f}{\partial y}(0, 0) = 0$. So, (C) is FALSE.

(D) Differentiability at (0,0):

A function is differentiable at a point if all partial derivatives exist AND the limit of the error term goes to zero. The condition for differentiability is:

$$\lim_{(h,k) \rightarrow (0,0)} \frac{f(h, k) - f(0, 0) - \frac{\partial f}{\partial x}(0, 0)h - \frac{\partial f}{\partial y}(0, 0)k}{\sqrt{h^2 + k^2}} = 0$$

Substituting the values we found:

$$\begin{aligned} \lim_{(h,k) \rightarrow (0,0)} \frac{f(h, k) - 0 - 1 \cdot h - 0 \cdot k}{\sqrt{h^2 + k^2}} &= \lim_{(h,k) \rightarrow (0,0)} \frac{f(h, k) - h}{\sqrt{h^2 + k^2}} \\ &= \lim_{(h,k) \rightarrow (0,0)} \frac{\frac{h}{|h|} \sqrt{h^2 + k^2} - h}{\sqrt{h^2 + k^2}} = \lim_{(h,k) \rightarrow (0,0)} \left(\frac{h}{|h|} - \frac{h}{\sqrt{h^2 + k^2}} \right) \end{aligned}$$

Let's check the limit along two paths.

Path 1: $k = 0, h \rightarrow 0^+$. Here $|h| = h$. The expression is $\lim_{h \rightarrow 0^+} (1 - \frac{h}{\sqrt{h^2}}) = 1 - 1 = 0$.

Path 2: $k = 0, h \rightarrow 0^-$. Here $|h| = -h$. The expression is $\lim_{h \rightarrow 0^-} (-1 - \frac{h}{\sqrt{h^2}}) = \lim_{h \rightarrow 0^-} (-1 - \frac{h}{-h}) = -1 - (-1) = 0$. This path doesn't show non-differentiability. Let's try another path.

Path 3: $h \rightarrow 0^-$ and $k = h$. The expression becomes

$$\lim_{h \rightarrow 0^-} \left(-1 - \frac{h}{\sqrt{h^2 + h^2}} \right) = \lim_{h \rightarrow 0^-} \left(-1 - \frac{h}{\sqrt{2}|h|} \right) = \lim_{h \rightarrow 0^-} \left(-1 - \frac{h}{\sqrt{2}(-h)} \right) = -1 + \frac{1}{\sqrt{2}} \neq 0$$

Since the limit is not zero, the function is not differentiable at (0,0). So, (D) is FALSE.

Step 4: Final Answer:

The true statements are (A) and (B).

 Quick Tip

For a function to be differentiable at a point, it's not enough for all directional derivatives to exist. The directional derivatives must also be a linear function of the direction vector components. Here $D_u f(0, 0)$ is not a linear function of u_1, u_2 , which is a strong hint that the function is not differentiable.

23. Let $\mathbb{Z}_{11}$ be the ring of integers modulo 11. Let $\mathbb{Z}_{11}[x]$ be the ring of all polynomials with coefficients in $\mathbb{Z}_{11}$. Which of the following statements is/are TRUE?

- (A) The polynomial $x^2 + x + 4$ is irreducible over $\mathbb{Z}_{11}$.
- (B) $\mathbb{Z}_{11}[x]/\langle x^2 + x + 4 \rangle$ is a field.
- (C) The ideal $\langle x^2 + x + 4 \rangle$ is maximal.
- (D) The ideal $\langle x^2 + x + 4 \rangle$ is not a prime ideal.

Correct Answer: (A) The polynomial $x^2 + x + 4$ is irreducible over $\mathbb{Z}_{11}$., (B) $\mathbb{Z}_{11}[x]/\langle x^2 + x + 4 \rangle$ is a field., (C) The ideal $\langle x^2 + x + 4 \rangle$ is maximal.

Solution:

Step 1: Understanding the Question:

We need to analyze the properties of the polynomial $p(x) = x^2 + x + 4$ over the field $\mathbb{Z}_{11}$. The statements concern its irreducibility and the properties of the ideal it generates.

Step 2: Key Formula or Approach:

1. **Irreducibility:** A quadratic polynomial $ax^2 + bx + c$ over a field F is reducible if and only if it has a root in F . We will check if $p(x)$ has any roots in $\mathbb{Z}_{11}$.
2. **Ring Theory Connections:** In a Principal Ideal Domain (PID) like $F[x]$ (where F is a field), the following are equivalent for a non-zero ideal $I = \langle p(x) \rangle$:
 - I is a maximal ideal.
 - I is a prime ideal.
 - The polynomial $p(x)$ is irreducible over F .
 - The quotient ring $F[x]/I$ is a field.

Therefore, if we can prove one of (A), (B), or (C), the others will follow. If we disprove one, the others are also disproved.

Step 3: Detailed Explanation:

(A) Irreducibility of $p(x) = x^2 + x + 4$:

We check for roots of $p(x)$ in $\mathbb{Z}_{11} = \{0, 1, 2, \dots, 10\}$. All calculations are modulo 11.

- $p(0) = 0^2 + 0 + 4 = 4 \neq 0$
- $p(1) = 1^2 + 1 + 4 = 6 \neq 0$
- $p(2) = 2^2 + 2 + 4 = 4 + 2 + 4 = 10 \neq 0$

- $p(3) = 3^2 + 3 + 4 = 9 + 3 + 4 = 16 \equiv 5 \neq 0$
- $p(4) = 4^2 + 4 + 4 = 16 + 4 + 4 = 24 \equiv 2 \neq 0$
- $p(5) = 5^2 + 5 + 4 = 25 + 5 + 4 = 34 \equiv 1 \neq 0$
- $p(6) = 6^2 + 6 + 4 = 36 + 6 + 4 = 46 \equiv 2 \neq 0$
- $p(7) = 7^2 + 7 + 4 = 49 + 7 + 4 = 60 \equiv 5 \neq 0$
- $p(8) = 8^2 + 8 + 4 = 64 + 8 + 4 = 76 \equiv 10 \neq 0$
- $p(9) = 9^2 + 9 + 4 = 81 + 9 + 4 = 94 \equiv 6 \neq 0$
- $p(10) = 10^2 + 10 + 4 = 100 + 10 + 4 = 114 \equiv 4 \neq 0$

Since $p(x)$ has no roots in $\mathbb{Z}_{11}$ and it is a degree-2 polynomial, it is irreducible over $\mathbb{Z}_{11}$. So, (A) is TRUE.

(B), (C), (D) Consequences of Irreducibility:

Since $\mathbb{Z}_{11}$ is a field, the polynomial ring $\mathbb{Z}_{11}[x]$ is a PID. We have established that $p(x) = x^2 + x + 4$ is an irreducible element in this ring.

- Because $p(x)$ is irreducible, the ideal it generates, $I = \langle x^2 + x + 4 \rangle$, is a maximal ideal. So, (C) is TRUE.
- Because every maximal ideal in a commutative ring with unity is also a prime ideal, I is a prime ideal. This means statement (D), "The ideal is not a prime ideal", is FALSE.
- Because I is a maximal ideal, the quotient ring $\mathbb{Z}_{11}[x]/I$ is a field. So, (B) is TRUE.

Step 4: Final Answer:

The statements (A), (B), and (C) are all true and are logical equivalents of each other in this context. Statement (D) is false.

💡 Quick Tip

For polynomial rings over a field F , remember the powerful chain of equivalences: $p(x)$ is irreducible $\iff \langle p(x) \rangle$ is a maximal ideal $\iff \langle p(x) \rangle$ is a non-zero prime ideal $\iff F[x]/\langle p(x) \rangle$ is a field. Proving any one of these proves all the others.

24. Consider $\mathbb{R}$ with the topology

$$\tau = \{A \subseteq \mathbb{R} : A^c \text{ is finite}\} \cup \{\emptyset\}.$$

Which of the following statements is/are TRUE?

- (A) $(\mathbb{R}, \tau)$ is a Hausdorff space.
- (B) Any finite subset of $(\mathbb{R}, \tau)$ is closed.
- (C) $(\mathbb{R}, \tau)$ is compact.
- (D) $(\mathbb{R}, \tau)$ is connected.

Correct Answer: (B) Any finite subset of $(\mathbb{R}, \tau)$ is closed., (C) $(\mathbb{R}, \tau)$ is compact., (D) $(\mathbb{R}, \tau)$ is connected.

Solution:

Step 1: Understanding the Question:

We are given the cofinite topology on the set of real numbers $\mathbb{R}$. The open sets are the empty set and any set whose complement is finite. The closed sets are the empty set, the whole space $\mathbb{R}$, and all finite subsets of $\mathbb{R}$. We need to check several topological properties: Hausdorff, closedness of finite sets, compactness, and connectedness.

Step 3: Detailed Explanation:

(A) Is $(\mathbb{R}, \tau)$ Hausdorff?

A space is Hausdorff (T_2) if for any two distinct points x, y , there exist disjoint open sets U and V such that $x \in U$ and $y \in V$.

Let $x, y \in \mathbb{R}$ with $x \neq y$. Let U and V be any two non-empty open sets containing x and y respectively.

By definition of the cofinite topology, U^c and V^c are finite sets.

We want to check if $U \cap V = \emptyset$. This is equivalent to checking if $(U \cap V)^c = \mathbb{R}$.

By De Morgan's laws, $(U \cap V)^c = U^c \cup V^c$.

Since U^c and V^c are finite, their union $U^c \cup V^c$ is also a finite set.

The set $\mathbb{R}$ is infinite. A finite set cannot be equal to $\mathbb{R}$.

Therefore, $U \cap V$ cannot be empty. Any two non-empty open sets in the cofinite topology on an infinite set must intersect. Thus, the space is not Hausdorff. (A) is FALSE.

(B) Is any finite subset closed?

Let F be a finite subset of $\mathbb{R}$. To check if F is closed, we must check if its complement, F^c , is open.

By the definition of the topology τ , a set is open if its complement is finite.

The complement of F^c is $(F^c)^c = F$. Since F is finite, F^c is an open set by definition.

Therefore, any finite set F is a closed set. (B) is TRUE.

(C) Is $(\mathbb{R}, \tau)$ compact?

A space is compact if every open cover has a finite subcover. Let $\{U_\alpha\}_{\alpha \in I}$ be an open cover of $\mathbb{R}$. Let's pick any non-empty open set from the cover, say U_{α_0} . By definition, $U_{\alpha_0}^c$ is a finite set. Let $U_{\alpha_0}^c = \{x_1, x_2, \dots, x_n\}$.

Since $\{U_\alpha\}$ is a cover, for each $x_i \in U_{\alpha_0}^c$, there must be an open set U_{α_i} in the cover such that $x_i \in U_{\alpha_i}$.

Now consider the collection $\{U_{\alpha_0}, U_{\alpha_1}, \dots, U_{\alpha_n}\}$. This is a finite collection of sets from the original cover.

Any point $x \in \mathbb{R}$ is either in U_{α_0} or in $U_{\alpha_0}^c$.

If $x \in U_{\alpha_0}$, it's covered.

If $x \in U_{\alpha_0}^c$, then x is one of the x_i , and so it is covered by U_{α_i} .

Thus, $\{U_{\alpha_0}, U_{\alpha_1}, \dots, U_{\alpha_n}\}$ is a finite subcover.

Since every open cover has a finite subcover, the space is compact. (C) is TRUE.

(D) Is $(\mathbb{R}, \tau)$ connected?

A space is disconnected if it can be written as the union of two disjoint, non-empty open sets.

Suppose $\mathbb{R} = U \cup V$, where U, V are open, non-empty, and $U \cap V = \emptyset$.

Since U and V are non-empty open sets, their complements U^c and V^c must be finite.

From $\mathbb{R} = U \cup V$ and $U \cap V = \emptyset$, we have $U = V^c$ and $V = U^c$.

This means U must be a finite set (since $U = V^c$), and V must be a finite set (since $V = U^c$).

But U and V are also non-empty open sets. For a non-empty set to be open in the cofinite topology, its complement must be finite. So U^c must be finite.

This leads to a contradiction: $V = U^c$ is finite, and U is finite, so their union $\mathbb{R} = U \cup V$ must be a finite set. But $\mathbb{R}$ is infinite.

The only way to avoid this contradiction is if one of the sets, say U , is empty. But we assumed they were non-empty.

Therefore, $\mathbb{R}$ cannot be written as a union of two disjoint non-empty open sets. It is connected.

(D) is TRUE.

Step 4: Final Answer:

The true statements are (B), (C), and (D).

💡 Quick Tip

The cofinite topology on an infinite set X is a classic example in topology. It is always T_1 (finite sets are closed), compact, and connected. However, it is never Hausdorff (T_2). Remembering these key properties can save a lot of time.

25. Consider the following partial differential equation:

$$a \frac{\partial^2 f(x, y)}{\partial x^2} + b \frac{\partial^2 f(x, y)}{\partial y^2} = 8f(x, y)$$

where a and b are distinct positive real numbers.

The combination(s) of the values of the real parameters ξ and η for which $f(x, y) = e^{\xi x + \eta y}$ is a solution of the given partial differential equation, is/are

- (A) $\xi = \frac{1}{\sqrt{a}}, \eta = \frac{2}{\sqrt{b}}$
- (B) $\xi = 0, \eta = 0$
- (C) $\xi = \frac{1}{\sqrt{2a}}, \eta = \frac{1}{\sqrt{2b}}$
- (D) $\xi = 0, \eta = 2\sqrt{\frac{2}{b}}$

Correct Answer: (A, D)

Solution:

Step 1: Understanding the Question:

We are given a second-order linear homogeneous partial differential equation (PDE). We need to find the values of parameters ξ and η for which the function $f(x, y) = e^{\xi x + \eta y}$ is a solution to this PDE. This involves substituting the function and its partial derivatives into the PDE and solving for ξ and η .

Step 2: Key Formula or Approach:

To check if $f(x, y) = e^{\xi x + \eta y}$ is a solution, we first need to compute its second partial derivatives with respect to x and y .

First partial derivatives:

$$\begin{aligned}\frac{\partial f}{\partial x} &= \xi e^{\xi x + \eta y} \\ \frac{\partial f}{\partial y} &= \eta e^{\xi x + \eta y}\end{aligned}$$

Second partial derivatives:

$$\begin{aligned}\frac{\partial^2 f}{\partial x^2} &= \xi^2 e^{\xi x + \eta y} \\ \frac{\partial^2 f}{\partial y^2} &= \eta^2 e^{\xi x + \eta y}\end{aligned}$$

Now, substitute these into the given PDE:

$$a(\xi^2 e^{\xi x + \eta y}) + b(\eta^2 e^{\xi x + \eta y}) = 8(e^{\xi x + \eta y})$$

Since $e^{\xi x + \eta y}$ is never zero, we can divide the entire equation by it:

$$a\xi^2 + b\eta^2 = 8$$

This is the characteristic equation that the parameters ξ and η must satisfy.

Step 3: Detailed Explanation:

We will check each option to see if it satisfies the condition $a\xi^2 + b\eta^2 = 8$.

For option (A): $\xi = \frac{1}{\sqrt{a}}, \eta = \frac{2}{\sqrt{b}}$

Substituting these values:

$$a \left(\frac{1}{\sqrt{a}} \right)^2 + b \left(\frac{2}{\sqrt{b}} \right)^2 = a \left(\frac{1}{a} \right) + b \left(\frac{4}{b} \right) = 1 + 4 = 5$$

Since $5 \neq 8$, option (A) as written is not a correct combination. However, as per the official answer key, this option is marked correct. This implies a typo in the question, either in the constant on the RHS of the PDE (it should have been 5) or in the values of ξ, η in this option. Assuming the answer key is correct, we proceed by noting this discrepancy.

For option (B): $\xi = 0, \eta = 0$

Substituting these values:

$$a(0)^2 + b(0)^2 = 0$$

Since $0 \neq 8$, this option is incorrect.

For option (C): $\xi = \frac{1}{\sqrt{2a}}, \eta = \frac{1}{\sqrt{2b}}$

Substituting these values:

$$a \left(\frac{1}{\sqrt{2a}} \right)^2 + b \left(\frac{1}{\sqrt{2b}} \right)^2 = a \left(\frac{1}{2a} \right) + b \left(\frac{1}{2b} \right) = \frac{1}{2} + \frac{1}{2} = 1$$

Since $1 \neq 8$, this option is incorrect.

For option (D): $\xi = 0, \eta = 2\sqrt{\frac{2}{b}}$

Substituting these values:

$$a(0)^2 + b \left(2\sqrt{\frac{2}{b}} \right)^2 = 0 + b \left(4 \cdot \frac{2}{b} \right) = b \left(\frac{8}{b} \right) = 8$$

Since $8 = 8$, this option is correct.

Step 4: Final Answer:

Based on direct calculation, only option (D) is correct. Following the official answer key which indicates both (A) and (D) are correct, we conclude there is likely a typo in the statement of option (A) or the PDE constant. For instance, if the RHS of the PDE was $5f(x, y)$, option (A) would be correct. Given the provided answers, we must select both.

💡 Quick Tip

When solving problems involving verifying solutions to PDEs, always derive the characteristic equation first. Then, systematically test each option against this equation. If there's a mismatch with the provided answer key, double-check your derivation, and if it's correct, be aware of potential typos in the exam question, which can happen in competitive exams.

26. Consider the following assignment problem where X, Y, Z are tasks, P, Q, R are agents and the cost matrix is given by

	<i>X</i>	<i>Y</i>	<i>Z</i>
<i>P</i>	4	2	8
<i>Q</i>	2	3	7
<i>R</i>	3	1	6

Which of the following statements is/are TRUE for an optimal assignment?

- (A) The cost is 9.
- (B) Agent P is assigned to task Y.
- (C) Agent Q is assigned to task X.
- (D) Agent R is assigned to task Y.

Correct Answer: (B, C)

Solution:

Step 1: Understanding the Question:

This is an assignment problem. The goal is to assign each agent to exactly one task such that the total cost is minimized. The cost matrix is given. We need to find the optimal assignment and the minimum cost, and then evaluate the given statements.

The cost matrix is:

	<i>X</i>	<i>Y</i>	<i>Z</i>
<i>P</i>	4	2	8
<i>Q</i>	2	3	7
<i>R</i>	3	1	6

Step 2: Key Formula or Approach:

For a small 3x3 matrix, we can enumerate all possible assignments and calculate their total costs. There are $3! = 6$ possible assignments.

Alternatively, we can use the Hungarian method, but enumeration is faster here.

The possible assignments are:

1. P-X, Q-Y, R-Z: Cost = $4 + 3 + 6 = 13$
2. P-X, Q-Z, R-Y: Cost = $4 + 7 + 1 = 12$
3. P-Y, Q-X, R-Z: Cost = $2 + 2 + 6 = 10$
4. P-Y, Q-Z, R-X: Cost = $2 + 7 + 3 = 12$
5. P-Z, Q-X, R-Y: Cost = $8 + 2 + 1 = 11$
6. P-Z, Q-Y, R-X: Cost = $8 + 3 + 3 = 14$

Step 3: Detailed Explanation:

By comparing the costs of all possible assignments, the minimum cost is 10.

This minimum cost corresponds to the assignment:

- Agent P is assigned to Task Y (Cost = 2).

- Agent Q is assigned to Task X (Cost = 2).
 - Agent R is assigned to Task Z (Cost = 6).
- Total Minimum Cost = 2 + 2 + 6 = 10.

Now let's evaluate the given statements based on this optimal assignment.

- (A) **The cost is 9.** This is FALSE. The minimum cost is 10.
- (B) **Agent P is assigned to task Y.** This is TRUE.
- (C) **Agent Q is assigned to task X.** This is TRUE.
- (D) **Agent R is assigned to task Y.** This is FALSE. Agent R is assigned to task Z.

Step 4: Final Answer:

The true statements are (B) and (C).

 Quick Tip

For small assignment problems (like 3x3 or 4x4), listing all permutations can be quicker and less error-prone than applying the full Hungarian algorithm. Always find the optimal assignment first before checking the options.

27. Consider the following differential equation:

$$2x(x-2)^2 \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + (x-2)y = 0$$

Which of the following statements is/are TRUE ?

- (A) The point $x = 2$ is not a singular point.
- (B) The point $x = 2$ is a singular point.
- (C) The point $x = 2$ is a regular singular point.
- (D) The point $x = 2$ is not a regular singular point.

Correct Answer: (B, D)

Solution:

Step 1: Understanding the Question:

We are asked to classify the point $x = 2$ for the given second-order linear ordinary differential equation. We need to determine if it is an ordinary point, a regular singular point, or an irregular singular point.

Step 2: Key Formula or Approach:

First, we write the differential equation in the standard form $y'' + P(x)y' + Q(x)y = 0$. Dividing the given equation by the coefficient of y'' , which is $2x(x-2)^2$, we get:

$$P(x) = \frac{3x}{2x(x-2)^2} = \frac{3}{2(x-2)^2}$$

$$Q(x) = \frac{x-2}{2x(x-2)^2} = \frac{1}{2x(x-2)}$$

A point $x = x_0$ is a singular point if either $P(x)$ or $Q(x)$ is not analytic (i.e., blows up) at $x = x_0$. Otherwise, it is an ordinary point.

A singular point x_0 is a regular singular point if both $(x - x_0)P(x)$ and $(x - x_0)^2Q(x)$ are analytic at $x = x_0$. If at least one of them is not analytic, the point is an irregular singular point.

Step 3: Detailed Explanation:

Check if $x=2$ is a singular point:

At $x = 2$, the denominator of $P(x)$, which is $2(x - 2)^2$, becomes zero. This means $P(x)$ is not analytic at $x = 2$.

Similarly, the denominator of $Q(x)$, which is $2x(x - 2)$, becomes zero at $x = 2$, so $Q(x)$ is not analytic at $x = 2$.

Since $P(x)$ is not analytic at $x = 2$, the point $x = 2$ is a singular point.

Therefore, statement (A) is FALSE and statement (B) is TRUE.

Check if $x=2$ is a regular singular point:

Now, we need to check the nature of this singularity. We test the analyticity of $(x - 2)P(x)$ and $(x - 2)^2Q(x)$ at $x = 2$.

1. For $(x - 2)P(x)$:

$$(x - 2)P(x) = (x - 2) \left(\frac{3}{2(x - 2)^2} \right) = \frac{3}{2(x - 2)}$$

This function is not analytic at $x = 2$ because it is undefined (approaches infinity).

Since one of the conditions for a regular singular point is violated (we don't even need to check the second one), the point $x = 2$ is an irregular singular point.

Therefore, statement (C) "The point $x = 2$ is a regular singular point" is FALSE.

Statement (D) "The point $x = 2$ is not a regular singular point" is TRUE.

Step 4: Final Answer:

The correct statements are (B) and (D).

💡 Quick Tip

To classify a point x_0 for a DE $A(x)y'' + B(x)y' + C(x)y = 0$:

1. x_0 is an ordinary point if $A(x_0) \neq 0$.

2. x_0 is a singular point if $A(x_0) = 0$.

3. To check if a singular point x_0 is regular, check the limits: $\lim_{x \rightarrow x_0} (x - x_0) \frac{B(x)}{A(x)}$ and $\lim_{x \rightarrow x_0} (x - x_0)^2 \frac{C(x)}{A(x)}$. If both limits are finite, it is a regular singular point. Otherwise, it is irregular. This limit-based check is often faster.

28. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation which reflects every vector in $\mathbb{R}^3$ through a two-dimensional subspace of $\mathbb{R}^3$. Let $P \in \mathbb{R}^{3 \times 3}$ be the matrix representation of T using the basis $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$.

Then the value of $2 \times \text{trace}(P) - 3 \times \text{determinant}(P)$ is equal to _____. (answer in integer)

Correct Answer: 5

Solution:

Step 1: Understanding the Question:

The linear transformation T reflects vectors through a 2D subspace. This is a geometric transformation. The problem gives the basis for the matrix representation of T , which is the standard basis for $\mathbb{R}^3$. A reflection through a subspace (plane) has specific eigenvalues. Any vector in the plane of reflection is unchanged (eigenvalue 1), and any vector normal to the plane is reversed (eigenvalue -1).

Step 2: Key Properties of Reflection:

Let W be the two-dimensional subspace (the plane of reflection). The space $\mathbb{R}^3$ can be decomposed as a direct sum $W \oplus W^\perp$, where $W^\perp$ is the one-dimensional subspace orthogonal to W (the normal line).

- For any vector $w \in W$, $T(w) = w$. This means $\lambda_1 = 1$ and $\lambda_2 = 1$ are eigenvalues, with eigenvectors forming a basis for W . The geometric multiplicity of eigenvalue 1 is 2.
- For any vector $v \in W^\perp$, $T(v) = -v$. This means $\lambda_3 = -1$ is an eigenvalue, with its eigenvector spanning $W^\perp$. The geometric multiplicity of eigenvalue -1 is 1.

The eigenvalues of the transformation T (and its matrix representation P) are $\{1, 1, -1\}$.

Step 3: Detailed Explanation:

The trace and determinant of a matrix are related to its eigenvalues.

- The trace of a matrix is the sum of its eigenvalues.

$$\text{trace}(P) = \lambda_1 + \lambda_2 + \lambda_3 = 1 + 1 + (-1) = 1.$$

- The determinant of a matrix is the product of its eigenvalues.

$$\text{determinant}(P) = \lambda_1 \times \lambda_2 \times \lambda_3 = 1 \times 1 \times (-1) = -1.$$

Now we can compute the required expression:

$$\begin{aligned} 2 \times \text{trace}(P) - 3 \times \text{determinant}(P) &= 2 \times (1) - 3 \times (-1) \\ &= 2 - (-3) = 2 + 3 = 5 \end{aligned}$$

Step 4: Final Answer:

The value of the expression is 5.

 Quick Tip

For geometric transformations like reflections and projections, understanding their effect on basis vectors is key. The trace and determinant are invariant under a change of basis. Using the properties of eigenvalues simplifies the calculation, avoiding the need to construct the matrix P explicitly for a general plane. For any reflection across a hyperplane in $\mathbb{R}^n$, the eigenvalues are $n - 1$ ones and a single minus one.

29. Let $x^3 - x + 1 \in \mathbb{Z}_3[x]$, where $\mathbb{Z}_3[x]$ is the ring of all polynomials with coefficients in $\mathbb{Z}_3$. Then the degree of the field extension $\mathbb{Z}_3[x]/\langle x^3 - x + 1 \rangle$ of $\mathbb{Z}_3$ is equal to _____. (answer in integer)

Correct Answer: 3

Solution:

Step 1: Understanding the Question:

We need to find the degree of a field extension. The extension is formed by taking the quotient of the polynomial ring $\mathbb{Z}_3[x]$ by the ideal generated by the polynomial $p(x) = x^3 - x + 1$. The base field is $\mathbb{Z}_3$.

Step 2: Key Formula or Approach:

The degree of the field extension $[\mathbb{F}[x]/\langle p(x) \rangle : \mathbb{F}]$, where $\mathbb{F}$ is a field and $p(x)$ is a polynomial in $\mathbb{F}[x]$, is equal to the degree of the polynomial $p(x)$, provided that $p(x)$ is irreducible over $\mathbb{F}$. So, the first step is to check if the polynomial $p(x) = x^3 - x + 1$ is irreducible over the field $\mathbb{Z}_3$.

Step 3: Detailed Explanation:

The field is $\mathbb{Z}_3 = \{0, 1, 2\}$.

The polynomial is $p(x) = x^3 - x + 1$.

A polynomial of degree 2 or 3 is irreducible over a field $\mathbb{F}$ if and only if it has no roots in $\mathbb{F}$.

We will test each element of $\mathbb{Z}_3$ to see if it is a root of $p(x)$. Remember that all calculations are performed modulo 3.

- For $x = 0$:

$$p(0) = (0)^3 - (0) + 1 = 1 \pmod{3} \neq 0$$

- For $x = 1$:

$$p(1) = (1)^3 - (1) + 1 = 1 - 1 + 1 = 1 \pmod{3} \neq 0$$

- For $x = 2$:

$$\begin{aligned} p(2) &= (2)^3 - (2) + 1 = 8 - 2 + 1 = 7 \\ 7 \pmod{3} &= 1 \neq 0 \end{aligned}$$

Since $p(x)$ has no roots in $\mathbb{Z}_3$, it is irreducible over $\mathbb{Z}_3$.

Because $p(x)$ is irreducible, the quotient ring $\mathbb{Z}_3[x]/\langle p(x) \rangle$ is a field extension of $\mathbb{Z}_3$.

The degree of this extension is the degree of the irreducible polynomial $p(x)$.

The degree of $p(x) = x^3 - x + 1$ is 3.

Step 4: Final Answer:

The degree of the field extension is 3.

 Quick Tip

To check for irreducibility of a low-degree polynomial (degree 2 or 3) over a finite field, the root test is the most efficient method. If the polynomial has no roots in the field, it's irreducible. For higher degrees, this test is not sufficient.

30. Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Using the modified Euler's method, the second approximation to $y(h)$, where $h = 0.05$ (step size), is equal to _____. (rounded off to TWO decimal places)

Correct Answer: 1.05

Solution:

Step 1: Understanding the Question:

We are asked to find an approximate value of $y(0.05)$ for the initial value problem $y' = x + y$, $y(0) = 1$, using the modified Euler's method. The term "second approximation" usually refers to an iterative application of the corrector step.

Step 2: Key Formula or Approach:

The modified Euler's method (also known as Heun's method or the predictor-corrector method) is given by:

Predictor step (Euler's method): $y_{n+1}^{(0)} = y_n + hf(x_n, y_n)$

Corrector step: $y_{n+1}^{(k+1)} = y_n + \frac{h}{2}[f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(k)})]$ for $k = 0, 1, 2, \dots$

Here, we have:

$$f(x, y) = x + y$$

$$x_0 = 0, y_0 = 1$$

$$h = 0.05, \text{ so } x_1 = x_0 + h = 0.05$$

We need to find $y_1 \approx y(0.05)$.

Step 3: Detailed Explanation:

First Approximation (Predictor):

This is the result from the standard Euler's method. Let's call it $y_1^{(0)}$.

$$y_1^{(0)} = y_0 + hf(x_0, y_0) = 1 + 0.05 \times f(0, 1) = 1 + 0.05 \times (0 + 1) = 1.05$$

Second Approximation (First Correction):

Now we apply the corrector formula once with $k = 0$ to get $y_1^{(1)}$.

$$y_1^{(1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(0)})]$$

$$y_1^{(1)} = 1 + \frac{0.05}{2}[f(0, 1) + f(0.05, 1.05)]$$

$$y_1^{(1)} = 1 + 0.025[(0 + 1) + (0.05 + 1.05)]$$

$$y_1^{(1)} = 1 + 0.025[1 + 1.10] = 1 + 0.025 \times 2.1 = 1 + 0.0525 = 1.0525$$

Typically, "Modified Euler's Method" refers to a single application of the predictor and corrector, which would give 1.0525. If "second approximation" means a second application of the corrector:

Third Approximation (Second Correction):

We apply the corrector formula again with $k = 1$ to get $y_1^{(2)}$.

$$y_1^{(2)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(1)})]$$

$$y_1^{(2)} = 1 + \frac{0.05}{2}[f(0, 1) + f(0.05, 1.0525)]$$

$$y_1^{(2)} = 1 + 0.025[(0 + 1) + (0.05 + 1.0525)]$$

$$y_1^{(2)} = 1 + 0.025[1 + 1.1025] = 1 + 0.025 \times 2.1025 = 1 + 0.0525625 = 1.0525625$$

The phrasing "second approximation" is slightly ambiguous. It could mean the result after one correction ($y_1^{(1)}$) or after two corrections ($y_1^{(2)}$). Both values, 1.0525 and 1.0525625, round to 1.05 when rounded to two decimal places.

Step 4: Final Answer:

Rounding the result to two decimal places:

$$1.0525 \approx 1.05$$

The value of the second approximation to $y(h)$ is 1.05.

💡 Quick Tip

The terminology for numerical methods can vary. "Modified Euler" might refer to the midpoint method or Heun's method (predictor-corrector). Both are second-order Runge-Kutta methods. For this problem, both interpretations lead to the same answer when rounded. Be familiar with the different forms of these methods.

31. Let

$$\alpha = \lim_{n \rightarrow \infty} \int_0^1 \frac{n^2 + (\sin(e^x))^n}{7n^2 + x^8} dx.$$

The value of 14α is equal to (answer in integer)

Correct Answer: 2

Solution:

Step 1: Understanding the Question:

We need to evaluate the limit of an integral. This often involves interchanging the limit and the integral, which requires justification, typically using a convergence theorem like the Dominated Convergence Theorem (DCT).

Step 2: Key Formula or Approach:

The Lebesgue Dominated Convergence Theorem states that if a sequence of functions $\{f_n\}$ converges pointwise almost everywhere to a function f , and there is an integrable function g such that $|f_n(x)| \leq g(x)$ for all n and for all x in the domain, then we can interchange the limit and integral:

$$\lim_{n \rightarrow \infty} \int f_n(x) dx = \int \lim_{n \rightarrow \infty} f_n(x) dx = \int f(x) dx$$

Let's apply this to our problem.

Let $f_n(x) = \frac{n^2 + (\sin(e^x))^n}{7n^2 + x^8}$ for $x \in [0, 1]$.

Step 3: Detailed Explanation:

1. Find the pointwise limit of $f_n(x)$:

Divide the numerator and denominator by n^2 :

$$f_n(x) = \frac{1 + \frac{(\sin(e^x))^n}{n^2}}{7 + \frac{x^8}{n^2}}$$

For $x \in [0, 1]$, $e^x \in [e^0, e^1] = [1, e]$. For any value in this range, $|\sin(e^x)| \leq 1$. Therefore, $|(\sin(e^x))^n| \leq 1^n = 1$.

As $n \rightarrow \infty$:

$$\frac{(\sin(e^x))^n}{n^2} \rightarrow 0 \quad (\text{since the numerator is bounded by 1})$$

$$\frac{x^8}{n^2} \rightarrow 0 \quad (\text{since } x \in [0, 1])$$

So, the pointwise limit is:

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \frac{1 + 0}{7 + 0} = \frac{1}{7}$$

2. Find a dominating function $g(x)$:

We need to find an integrable function $g(x)$ on $[0, 1]$ such that $|f_n(x)| \leq g(x)$ for all n .

$$|f_n(x)| = \left| \frac{n^2 + (\sin(e^x))^n}{7n^2 + x^8} \right|$$

Since $x \in [0, 1]$, $x^8 \geq 0$, so the denominator $7n^2 + x^8 \geq 7n^2$. For the numerator, using the triangle inequality: $|n^2 + (\sin(e^x))^n| \leq |n^2| + |(\sin(e^x))^n| \leq n^2 + 1$. So,

$$|f_n(x)| \leq \frac{n^2 + 1}{7n^2} = \frac{1}{7} + \frac{1}{7n^2}$$

For $n \geq 1$, we have $\frac{1}{7n^2} \leq \frac{1}{7}$. Thus,

$$|f_n(x)| \leq \frac{1}{7} + \frac{1}{7} = \frac{2}{7}$$

We can choose our dominating function to be the constant function $g(x) = 2/7$. The function $g(x) = 2/7$ is integrable on $[0, 1]$ since $\int_0^1 (2/7) dx = 2/7 < \infty$.

3. Apply the Dominated Convergence Theorem:

Since the conditions of the DCT are met, we can swap the limit and the integral.

$$\alpha = \int_0^1 \left(\lim_{n \rightarrow \infty} f_n(x) \right) dx = \int_0^1 \frac{1}{7} dx$$

$$\alpha = \left[\frac{x}{7} \right]_0^1 = \frac{1}{7} - 0 = \frac{1}{7}$$

4. Calculate the final value:

The question asks for the value of 14α .

$$14\alpha = 14 \times \frac{1}{7} = 2$$

Step 4: Final Answer:

The value of 14α is 2.

💡 Quick Tip

When faced with a limit of an integral, think of convergence theorems. The Dominated Convergence Theorem is very powerful. The main steps are: find the pointwise limit of the integrand, find a suitable integrable dominating function, and then apply the theorem to swap the limit and integral.

32. Let $X = (\mathbb{R}^3, \|\cdot\|_1)$ where $\left\| \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right\|_1 = |x| + |y| + |z|$ and $T : X \rightarrow X$ be a linear transformation defined as

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 2 & -2 \\ 1 & 3 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Then the operator norm of T is equal to _____. (answer in integer)

Correct Answer: 8

Solution:

Step 1: Understanding the Question:

We are asked to find the operator norm of a linear transformation T from $\mathbb{R}^3$ to $\mathbb{R}^3$, where the norm used in both the domain and codomain is the L_1 -norm (or taxicab norm). The transformation is given by a matrix.

Step 2: Key Formula or Approach:

The operator norm of a linear transformation $T : (\mathbb{R}^n, \|\cdot\|_p) \rightarrow (\mathbb{R}^m, \|\cdot\|_q)$ represented by a matrix A is denoted by $\|A\|_{p,q}$. When $p = q = 1$, the norm is often denoted as $\|A\|_1$. The formula for the operator norm $\|A\|_1$ (mapping from an L_1 space to an L_1 space) is the maximum of the L_1 -norms of the column vectors of the matrix A . If $A = [a_{ij}]$, then

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}|$$

This is also known as the maximum absolute column sum.

Step 3: Detailed Explanation:

The matrix representation of the linear transformation T is:

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 2 & -2 \\ 1 & 3 & -3 \end{pmatrix}$$

We need to calculate the sum of the absolute values of the elements in each column.

- **Column 1:**

$$\sum_{i=1}^3 |a_{i1}| = |1| + |2| + |1| = 1 + 2 + 1 = 4$$

- **Column 2:**

$$\sum_{i=1}^3 |a_{i2}| = |1| + |2| + |3| = 1 + 2 + 3 = 6$$

- **Column 3:**

$$\sum_{i=1}^3 |a_{i3}| = |3| + |-2| + |-3| = 3 + 2 + 3 = 8$$

The operator norm $\|T\|_1$ is the maximum of these column sums.

$$\|T\|_1 = \max\{4, 6, 8\} = 8$$

Step 4: Final Answer:

The operator norm of T is 8.

💡 Quick Tip

Remember the standard matrix norms:

- $\|A\|_1$ (max absolute column sum): useful for L_1 spaces.
 - $\|A\|_\infty$ (max absolute row sum): useful for L_∞ spaces.
 - $\|A\|_2$ (spectral norm, max singular value): useful for L_2 (Euclidean) spaces.
- Identifying the norm in the question ($\|\cdot\|_1$) is the crucial first step.

33. Consider the following Linear Programming Problem (LPP): maximize $z = 10x_1 + 20x_2$ subject to $x_1 \leq 36$ $x_2 \geq 42$ $x_1 + x_2 \geq 48$ $5x_1 + x_2 \leq 150$ $x_1, x_2 \geq 0$ The maximum value of z in the above LPP is equal to ----- . (answer in integer)

Correct Answer: 3000

Solution:

Step 1: Understanding the Question:

This is a two-variable Linear Programming Problem. We need to find the maximum value of the objective function z subject to a set of linear inequality constraints. The graphical method is suitable for solving this.

Step 2: Key Formula or Approach:

The graphical method involves the following steps:

1. Plot the boundary lines for each inequality constraint.
2. Identify the feasible region, which is the intersection of all regions satisfying the constraints.
3. Determine the corner points (vertices) of the feasible region.
4. Evaluate the objective function z at each corner point.
5. The maximum value of z will be the largest value found in the previous step.

Step 3: Detailed Explanation:

Let's find the corner points of the feasible region defined by the constraints.

The constraints are:

- (1) $x_1 \leq 36$
- (2) $x_2 \geq 42$
- (3) $x_1 + x_2 \geq 48$
- (4) $5x_1 + x_2 \leq 150$
- (5) $x_1 \geq 0, x_2 \geq 0$ (non-negativity)

Let's analyze the constraints on x_1 . From (4), since $x_2 \geq 42$, we have $5x_1 + 42 \leq 5x_1 + x_2 \leq 150$. This implies $5x_1 \leq 150 - 42 = 108$, so $x_1 \leq 21.6$.

The constraint $x_1 \leq 36$ is therefore redundant, as the feasible region is already limited to $x_1 \leq 21.6$.

The feasible region is determined by: $0 \leq x_1 \leq 21.6$, $x_2 \geq 42$, $x_1 + x_2 \geq 48$, and $5x_1 + x_2 \leq 150$.

Now we find the vertices by finding the intersection of the boundary lines.

Vertex A: Intersection of $x_1 + x_2 = 48$ and $x_2 = 42$. $x_1 + 42 = 48 \Rightarrow x_1 = 6$. So, A = (6, 42).

Check feasibility: All constraints are satisfied.

$$z(A) = 10(6) + 20(42) = 60 + 840 = 900.$$

Vertex B: Intersection of $5x_1 + x_2 = 150$ and $x_2 = 42$. $5x_1 + 42 = 150 \Rightarrow 5x_1 = 108 \Rightarrow x_1 = 21.6$. So, B = (21.6, 42).

Check feasibility: All constraints are satisfied.

$$z(B) = 10(21.6) + 20(42) = 216 + 840 = 1056.$$

Vertex C: Intersection of $x_1 = 0$ and $x_1 + x_2 = 48$. $x_2 = 48$. So, $C = (0, 48)$.
 Check feasibility: $0 \leq 36$, $48 \geq 42$, $0 + 48 \geq 48$, $5(0) + 48 \leq 150$. All satisfied.
 $z(C) = 10(0) + 20(48) = 960$.

Vertex D: Intersection of $x_1 = 0$ and $5x_1 + x_2 = 150$. $x_2 = 150$. So, $D = (0, 150)$.
 Check feasibility: $0 \leq 36$, $150 \geq 42$, $0 + 150 \geq 48$, $5(0) + 150 \leq 150$. All satisfied.
 $z(D) = 10(0) + 20(150) = 3000$.

The corner points of the feasible region are $(6, 42)$, $(21.6, 42)$, $(0, 150)$, and $(0, 48)$.
 The values of the objective function at these vertices are:

- $z(6, 42) = 900$
- $z(21.6, 42) = 1056$
- $z(0, 48) = 960$
- $z(0, 150) = 3000$

Step 4: Final Answer:

The maximum value of z is the largest of these values, which is 3000.

 Quick Tip

In graphical LPP, always check for redundant constraints first. This can simplify the process of finding the feasible region and its vertices. A constraint is redundant if the feasible region it defines is already contained within the region defined by other constraints. In this case, $x_1 \leq 36$ was redundant.

34. The number of distinct topologies on the set $\{1, 2, 3\}$ consisting of exactly four elements is equal to _____. (answer in integer)

Correct Answer: 9

Solution:

Step 1: Understanding the Question:

We need to find the number of distinct topologies on a set $X = \{1, 2, 3\}$ that have exactly 4 open sets.

A topology τ on X must satisfy three axioms:

1. The empty set $\emptyset$ and the whole set X are in τ .
2. The union of any collection of sets in τ is also in τ .
3. The intersection of any finite collection of sets in τ is also in τ .

Step 2: Key Formula or Approach:

Let the topology be $\tau = \{\emptyset, X, A, B\}$, where A and B are distinct, non-empty, proper subsets of X . For τ to be a topology, it must be closed under unions and intersections. This means:

- $A \cup B \in \tau$
- $A \cap B \in \tau$

We need to find all pairs of subsets $\{A, B\}$ of X that satisfy these closure properties.

Step 3: Detailed Explanation:

We analyze the possible relationships between the sets A and B .

Case 1: Nested Sets ($A \subset B$ or $B \subset A$)

Without loss of generality, let A be a proper subset of B .

Then $A \cup B = B$ and $A \cap B = A$.

Since both A and B are in τ , the closure axioms are satisfied. We need to count the number of such chains of subsets $\emptyset \subset A \subset B \subset X$.

This requires $|A| < |B|$. Since A and B are proper subsets, their sizes can be 1 or 2.

The only possibility is $|A| = 1$ and $|B| = 2$.

- **Choose A (a singleton set):** There are $\binom{3}{1} = 3$ ways to choose a singleton set (e.g., $\{1\}$).
- **Choose B (a doubleton set containing A):** For each choice of $A = \{a\}$, there are 2 remaining elements in X . We can form a set of size 2 by adding one of these two elements to A . So there are $\binom{2}{1} = 2$ ways (e.g., if $A = \{1\}$, B can be $\{1, 2\}$ or $\{1, 3\}$). The total number of such chains (and thus topologies) is $3 \times 2 = 6$. Example: $\tau = \{\emptyset, X, \{1\}, \{1, 2\}\}$ is a valid topology.

Case 2: Disjoint Sets ($A \cap B = \emptyset$)

In this case, $A \cap B = \emptyset$, which is in τ . So this condition is satisfied.

We also need $A \cup B \in \tau = \{\emptyset, X, A, B\}$.

- $A \cup B$ cannot be $\emptyset$ since A, B are non-empty.
- $A \cup B$ cannot be A or B since A, B are non-empty and disjoint.
- Therefore, we must have $A \cup B = X$.

This means that $\{A, B\}$ must be a partition of X into two non-empty sets.

We need to count the number of ways to partition the 3-element set X into two non-empty blocks. This is given by the Stirling number of the second kind, $S(3, 2)$.

A partition of $\{1, 2, 3\}$ into two blocks must consist of one singleton and one doubleton.

- Choose the singleton set: $\binom{3}{1} = 3$ ways. The other set is its complement.

The possible partitions are $\{\{1\}, \{2, 3\}\}$, $\{\{2\}, \{1, 3\}\}$, and $\{\{3\}, \{1, 2\}\}$.

Each partition gives rise to a valid topology. For example, $\tau = \{\emptyset, X, \{1\}, \{2, 3\}\}$.

This gives 3 more topologies.

Case 3: Overlapping, Non-nested Sets

Suppose $A \not\subset B$, $B \not\subset A$, and $A \cap B \neq \emptyset$.

Let $A = \{1, 2\}$ and $B = \{1, 3\}$.

Then $A \cap B = \{1\}$ and $A \cup B = \{1, 2, 3\} = X$.

For $\tau = \{\emptyset, X, A, B\}$ to be a topology, $A \cap B = \{1\}$ must be in τ . But it is not (since $A \cap B \neq \emptyset, X, A, B$). Thus, this case does not yield a valid topology of size 4.

Total Count:

Summing up the topologies from the valid cases:

Total = (Topologies from Case 1) + (Topologies from Case 2) = $6 + 3 = 9$.

Step 4: Final Answer:

There are 9 distinct topologies on the set $\{1, 2, 3\}$ with exactly four elements.

 Quick Tip

When counting topologies with a specific number of open sets, systematically consider the relationships (nested, disjoint, overlapping) between the non-trivial open sets. For each case, check if the union and intersection axioms force new sets into the topology, which would violate the size constraint.

35. For $a_n \in \mathbb{C}, n = 0, 1, 2, \dots$, the power series

$$\sum_{n=0}^{\infty} a_n (z - 2)^n$$

converges at $z = 5$ and diverges at $z = -1$. Then the radius of convergence of this power series is equal to _____. (answer in integer)

Correct Answer: 3

Solution:

Step 1: Understanding the Question:

We are given a power series centered at $z_0 = 2$. We are given information about its convergence at two specific points. We need to determine the radius of convergence, R .

Step 2: Key Formula or Approach:

A power series $\sum a_n (z - z_0)^n$ with radius of convergence R has the following properties: - It converges absolutely for all z such that $|z - z_0| < R$. - It diverges for all z such that $|z - z_0| > R$. - The behavior on the circle of convergence, $|z - z_0| = R$, can be convergence or divergence at different points.

Step 3: Detailed Explanation:

The center of the given power series is $z_0 = 2$.

Information from convergence at $z = 5$:

The series converges at $z = 5$. The distance of this point from the center is $|5 - 2| = 3$. According to the properties of a power series, the radius of convergence R must be at least this distance. This is because the region of convergence is a disk, and if a point is included, the entire open disk of that radius must be included. So, we must have $R \geq |5 - 2|$, which implies $R \geq 3$.

Information from divergence at $z = -1$:

The series diverges at $z = -1$. The distance of this point from the center is $|-1 - 2| = |-3| = 3$.

According to the properties of a power series, the radius of convergence R cannot be greater than this distance. If R were greater than 3, the series would have to converge at $z = -1$, which contradicts the given information. So, we must have $R \leq |-1 - 2|$, which implies $R \leq 3$.

Combining the information:

From the convergence information, we have $R \geq 3$. From the divergence information, we have $R \leq 3$. The only value of R that satisfies both inequalities is $R = 3$.

Step 4: Final Answer:

The radius of convergence of the power series is 3.

💡 Quick Tip

The radius of convergence R defines a "boundary". Convergence inside the circle $|z - z_0| < R$ implies R is at least that distance. Divergence outside the circle $|z - z_0| > R$ implies R is at most that distance. Information about points on the boundary itself provides the most precise constraint on R .

36. The solution $y(x) = \begin{pmatrix} y_1(x) \\ y_2(x) \end{pmatrix}$ of the initial value problem

$$\frac{dy}{dx} = \begin{pmatrix} -3 & 4 \\ -2 & 3 \end{pmatrix} y, \quad y(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

is equal to

- (A) $3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^x + \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-x}$
- (B) $-3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^x + \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-x}$
- (C) $3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^x + 2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-x}$
- (D) $3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^x - \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-x}$

Correct Answer: (D)

Solution:

Step 1: Understanding the Question:

We must solve a system of first-order linear differential equations with a given initial condition. This is an initial value problem (IVP).

Step 2: Key Formula or Approach:

The standard method is to find the eigenvalues (λ) and eigenvectors (v) of the coefficient matrix

$A = \begin{pmatrix} -3 & 4 \\ -2 & 3 \end{pmatrix}$. The general solution is $y(x) = c_1 v_1 e^{\lambda_1 x} + c_2 v_2 e^{\lambda_2 x}$. The constants c_1, c_2 are found using the initial condition.

Step 3: Detailed Explanation:

First, find the eigenvalues by solving $\det(A - \lambda I) = 0$:

$$(-3 - \lambda)(3 - \lambda) - (4)(-2) = \lambda^2 - 9 + 8 = \lambda^2 - 1 = 0 \implies \lambda_1 = 1, \lambda_2 = -1.$$

Next, find the corresponding eigenvectors:

For $\lambda_1 = 1$: $(A - I)v = 0 \implies \begin{pmatrix} -4 & 4 \\ -2 & 2 \end{pmatrix} v_1 = 0 \implies v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

For $\lambda_2 = -1$: $(A + I)v = 0 \implies \begin{pmatrix} -2 & 4 \\ -2 & 4 \end{pmatrix} v_2 = 0 \implies v_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

The general solution is $y(x) = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^x + c_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-x}$.

Apply the initial condition $y(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$:

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} \implies \begin{cases} c_1 + 2c_2 = 1 \\ c_1 + c_2 = 2 \end{cases}$$

Solving this system gives $c_2 = -1$ and $c_1 = 3$.

Step 4: Final Answer:

The particular solution is $y(x) = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^x - 1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{-x}$, which matches option (D).

 Quick Tip

For 2x2 systems, finding eigenvalues and eigenvectors is generally fast. The characteristic equation is $\lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$. Here, $\text{tr}(A) = 0, \det(A) = -1$, so $\lambda^2 - 1 = 0$. This is a quick check.

37. Let $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 16 \end{pmatrix}$. Define an inner product on $\mathbb{R}^3$ with respect to P as

$\langle x, y \rangle_P = x^T P y$, for all $x, y \in \mathbb{R}^3$. Consider the subspace $V = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$.

Which one of the following sets is an orthonormal basis of V with respect to $\langle x, y \rangle_P$?

(A) $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$

$$(B) \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$(C) \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1/3 \\ 1/4 \end{pmatrix} \right\}$$

$$(D) \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \right\}$$

Correct Answer: (C)

Solution:

Step 1: Understanding the Question:

We must find an orthonormal basis for the subspace V with respect to a given non-standard inner product defined by the matrix P . This requires using the Gram-Schmidt process.

Step 2: Key Formula or Approach:

Let the basis vectors for V be $v_1 = (1, 0, 0)^T$ and $v_2 = (1, 1, 1)^T$. We apply the Gram-Schmidt process. 1. First vector: $e_1 = \frac{v_1}{\|v_1\|_P}$, where $\|v_1\|_P^2 = \langle v_1, v_1 \rangle_P = v_1^T P v_1$. 2. Second vector: $u_2 = v_2 - \langle v_2, e_1 \rangle_P e_1$, and then normalize it: $e_2 = \frac{u_2}{\|u_2\|_P}$.

Step 3: Detailed Explanation:

Let's find the first vector e_1 :

$$\|v_1\|_P^2 = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 16 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 1. \text{ So, } \|v_1\|_P = 1, \text{ and } e_1 = v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

Now, find the second orthogonal vector u_2 :

$$\langle v_2, e_1 \rangle_P = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 16 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 1.$$

$$u_2 = v_2 - \langle v_2, e_1 \rangle_P e_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - 1 \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Finally, normalize u_2 :

$$\|u_2\|_P^2 = \begin{pmatrix} 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 16 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 9(1)^2 + 16(1)^2 = 25.$$

$$\|u_2\|_P = 5. \text{ So, } e_2 = \frac{u_2}{5} = \begin{pmatrix} 0 \\ 1/5 \\ 1/5 \end{pmatrix}.$$

The correct orthonormal basis is $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1/5 \\ 1/5 \end{pmatrix} \right\}$. This is not among the options, indicating a flawed question.

Revisiting the problem based on the answer key: The official key is (C). Let's check the properties of the vectors in (C): $w_1 = (1, 0, 0)^T$, $w_2 = (0, 1/3, 1/4)^T$. - $\langle w_1, w_2 \rangle_P = (1 \ 0 \ 0) P w_2 = 0$. They are orthogonal. - $\|w_1\|_P = 1$. It is normalized. - $\|w_2\|_P^2 = (0 \ 1/3 \ 1/4) P w_2 = 9(1/3)^2 + 16(1/4)^2 = 1 + 1 = 2$. It is not normalized. - Also, w_2 is not in the span of V since any vector in V of the form $(0, a, b)^T$ must have $a = b$. The question is definitively flawed. Option (C) provides an orthogonal, but not orthonormal, set. The second vector is not in the subspace V . Given the constraints, we select (C) as it is the only option presenting an orthogonal pair (besides the first vector).

Step 4: Final Answer:

The problem statement is inconsistent. Based on the provided options and answer key, we choose (C), acknowledging that the set is only orthogonal and not fully orthonormal, and the second vector is not in the specified subspace.

💡 Quick Tip

In competitive exams, you might encounter flawed questions. If your rigorous derivation leads to a result not in the options, re-check your work. If the conflict persists, analyze the options for partial correctness (e.g., orthogonality instead of orthonormality) to guess the intended answer.

38. The value of the contour integral

$$\oint_{\Gamma} \frac{(1-z)(3\cos z + 5\sin z)}{1-z^{101}} dz,$$

where $\Gamma = \{z \in \mathbb{C} : |z| = 3\}$, oriented in the counter-clockwise direction, is equal to

- (A) $2\pi i$
- (B) $-2\pi i$
- (C) 0
- (D) $4\pi i$

Correct Answer: (C)

Solution:

Step 1: Understanding the Question:

We are asked to compute a contour integral. The contour encloses all singularities of the integrand, which suggests using the residue at infinity.

Step 2: Key Formula or Approach:

First, simplify the integrand $f(z)$. For $z \neq 1$, $1 - z^{101} = (1 - z)(1 + z + \cdots + z^{100})$. So,

$f(z) = \frac{3 \cos z + 5 \sin z}{1 + z + z^2 + \dots + z^{100}}$. The poles are the roots of $1 + z + \dots + z^{100} = 0$, which are the 101st roots of unity except for $z = 1$. All these poles have modulus 1 and are thus inside the contour $|z| = 3$.

The integral can be computed as $\oint_{\Gamma} f(z) dz = -2\pi i \operatorname{Res}(f, \infty)$. The residue at infinity is given by $\operatorname{Res}(f, \infty) = -\operatorname{Res}_{w=0} \left[\frac{1}{w^2} f\left(\frac{1}{w}\right) \right]$.

Step 3: Detailed Explanation:

Let's find the expression for $\frac{1}{w^2} f\left(\frac{1}{w}\right)$.

$$f\left(\frac{1}{w}\right) = \frac{3 \cos(1/w) + 5 \sin(1/w)}{1 + 1/w + \dots + 1/w^{100}} = \frac{w^{100}(3 \cos(1/w) + 5 \sin(1/w))}{w^{100} + w^{99} + \dots + 1}.$$

$$-\frac{1}{w^2} f\left(\frac{1}{w}\right) = -\frac{w^{98}(3 \cos(1/w) + 5 \sin(1/w))}{1 + w + \dots + w^{100}}.$$

To find the residue at $w = 0$, we need the coefficient of the w^{-1} term in the Laurent expansion.

As $w \rightarrow 0$, the denominator approaches 1. Let's expand the numerator:

$$3 \cos(1/w) + 5 \sin(1/w) = 3\left(1 - \frac{1}{2!w^2} + \dots\right) + 5\left(\frac{1}{w} - \frac{1}{3!w^3} + \dots\right) = 3 + \frac{5}{w} - \frac{3}{2w^2} + \dots$$

$$\text{So, } -w^{98}(3 \cos(1/w) + 5 \sin(1/w)) = -w^{98}\left(3 + \frac{5}{w} - \frac{3}{2w^2} + \dots\right) = -3w^{98} - 5w^{97} + \frac{3}{2}w^{96} - \dots$$

This expression is a power series in w with only non-negative powers.

It is analytic at $w = 0$.

The coefficient of w^{-1} is 0. Thus, the residue at $w = 0$ is 0.

Step 4: Final Answer:

Since $\operatorname{Res}_{w=0} \left[-\frac{1}{w^2} f\left(\frac{1}{w}\right) \right] = 0$, we have $\operatorname{Res}(f, \infty) = 0$. The integral is $\oint_{\Gamma} f(z) dz = -2\pi i \operatorname{Res}(f, \infty) = 0$.

Quick Tip

When a contour encloses all the finite singularities of a function, the integral is determined solely by the residue at infinity. This can be much simpler than summing a large number of residues.

39. Let $f(z) = \sum_{n=1}^{\infty} 5^{-n} \cos(nz)$. On which one of the following domains, does $f(z)$ represent an analytic function?

- (A) $\{z \in \mathbb{C} : |\operatorname{Im} z| < \ln 5\}$
- (B) $\{z \in \mathbb{C} : |\operatorname{Im} z| > \ln 5\}$
- (C) $\{z \in \mathbb{C} : |\operatorname{Re} z| < \ln 5\}$
- (D) $\{z \in \mathbb{C} : |\operatorname{Re} z| > \ln 5\}$

Correct Answer: (A)

Solution:

Step 1: Understanding the Question:

We need to find the domain of convergence for a given series of complex functions. A uniformly convergent series of analytic functions is analytic.

Step 2: Key Formula or Approach:

We express $\cos(nz)$ using Euler's formula: $\cos(nz) = \frac{e^{inz} + e^{-inz}}{2}$. This transforms the series into a sum of two geometric series.

$$f(z) = \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{e^{iz}}{5} \right)^n + \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{e^{-iz}}{5} \right)^n.$$

A geometric series $\sum r^n$ converges if and only if $|r| < 1$.

Step 3: Detailed Explanation:

Let $z = x + iy$. We need both series to converge.

Condition 1: For the first series, $\left| \frac{e^{iz}}{5} \right| < 1$.

$$|e^{iz}| = |e^{i(x+iy)}| = |e^{-y+ix}| = |e^{-y}| |e^{ix}| = e^{-y}.$$

$$\text{So, } e^{-y}/5 < 1 \implies e^{-y} < 5 \implies -y < \ln 5 \implies y > -\ln 5.$$

Condition 2: For the second series, $\left| \frac{e^{-iz}}{5} \right| < 1$.

$$|e^{-iz}| = |e^{-i(x+iy)}| = |e^{y-ix}| = |e^y| |e^{-ix}| = e^y.$$

$$\text{So, } e^y/5 < 1 \implies e^y < 5 \implies y < \ln 5.$$

For $f(z)$ to converge, both conditions must hold: $-\ln 5 < y < \ln 5$.

Step 4: Final Answer:

The domain of convergence is where $-\ln 5 < \text{Im}(z) < \ln 5$, which is equivalent to $|\text{Im}(z)| < \ln 5$. This is option (A).

💡 Quick Tip

For series with $\cos(nz)$ or $\sin(nz)$, converting to complex exponentials is a powerful strategy. It often simplifies the problem to analyzing geometric series, for which the convergence condition is straightforward.

40. Let $\delta(t)$ be the unit impulse function defined by $\delta(x - x_0) = 0$ when $x \neq x_0$ and $\int_{-\infty}^{\infty} \delta(x - x_0) dx = 1$. Consider the following initial value problem

$$\frac{d^2 y}{dx^2} + \frac{dy}{dx} + 2y = \delta(x - 5)$$

with $y(0) = 0$ and $\frac{dy}{dx}(0) = 0$. Which one of the following is TRUE?

- (A) $y(10) = \frac{10}{\sqrt{15}} e^{-5/4} \sin\left(\frac{\sqrt{15}}{4}\right)$
 (B) $y(10) = \frac{2}{\sqrt{15}} e^{-5/4} \sin\left(\frac{5\sqrt{15}}{4}\right)$
 (C) $y(10) = \frac{10}{\sqrt{15}} e^{5/4} \sin\left(\frac{\sqrt{15}}{4}\right)$

$$(D) \ y(10) = \frac{2}{\sqrt{15}} e^{5/4} \sin\left(\frac{5\sqrt{15}}{4}\right)$$

Correct Answer: (B)

Solution:

Step 1: Understanding the Question:

We must solve a second-order ODE with an impulse forcing function using Laplace transforms. The options contain terms like $\sqrt{15}$ and $e^{-5/4}$, which suggests the characteristic roots are complex.

Step 2: Key Formula or Approach:

The given ODE $y'' + y' + 2y = \delta(x - 5)$ has characteristic equation $s^2 + s + 2 = 0$ with roots $s = -1/2 \pm i\sqrt{7}/2$. The terms in the options $(\sqrt{15}/4, -1/4)$ correspond to roots $-1/4 \pm i\sqrt{15}/4$, which arise from the characteristic equation $2s^2 + s + 2 = 0$. We assume the intended ODE was $2y'' + y' + 2y = \delta(x - 5)$.

We apply the Laplace transform $\mathcal{L}$ with initial conditions $y(0) = y'(0) = 0$.

$$\mathcal{L}\{y'\} = sY(s), \mathcal{L}\{y''\} = s^2Y(s), \mathcal{L}\{\delta(x - 5)\} = e^{-5s}.$$

Step 3: Detailed Explanation:

Transforming the corrected ODE $2y'' + y' + 2y = \delta(x - 5)$:

$$2s^2Y(s) + sY(s) + 2Y(s) = e^{-5s} \implies Y(s) = \frac{e^{-5s}}{2s^2 + s + 2}.$$

To find the inverse Laplace transform, we rewrite the denominator:

$$Y(s) = \frac{1}{2} \frac{e^{-5s}}{s^2 + s/2 + 1} = \frac{1}{2} \frac{e^{-5s}}{(s+1/4)^2 + 15/16}.$$

Let $H(s) = \frac{1}{(s+1/4)^2 + (\sqrt{15}/4)^2}$. We know $\mathcal{L}^{-1}\{H(s)\} = \frac{4}{\sqrt{15}} e^{-x/4} \sin\left(\frac{\sqrt{15}}{4}x\right)$. So, $y(x) =$

$$\mathcal{L}^{-1}\left\{\frac{1}{2}e^{-5s}H(s)\right\} = \frac{1}{2} \cdot \frac{4}{\sqrt{15}} e^{-(x-5)/4} \sin\left(\frac{\sqrt{15}}{4}(x-5)\right) u(x-5).$$

$$y(x) = \frac{2}{\sqrt{15}} e^{-(x-5)/4} \sin\left(\frac{\sqrt{15}}{4}(x-5)\right) u(x-5).$$

Now, evaluate at $x = 10$:

$$y(10) = \frac{2}{\sqrt{15}} e^{-(10-5)/4} \sin\left(\frac{\sqrt{15}}{4}(10-5)\right) = \frac{2}{\sqrt{15}} e^{-5/4} \sin\left(\frac{5\sqrt{15}}{4}\right).$$

Step 4: Final Answer:

The result matches option (B). The original question likely had a typo.

💡 Quick Tip

If the form of the answer options (e.g., arguments of sines, exponential terms) suggests a different characteristic equation than the one given, it is highly probable there is a typo in the question. Work backwards from the options to infer the intended equation.

41. Which one of the following statements is TRUE?

- (A) The set of all 2×2 real matrices under usual matrix addition and multiplication is a Unique Factorization Domain.
- (B) The set of all real valued functions under usual function addition and multiplication is a Unique Factorization Domain.
- (C) The set of all polynomials with coefficients in $\mathbb{R}$ under usual polynomial addition and multiplication is a Unique Factorization Domain.
- (D) The set of Riemann integrable functions in $[0, 1]$ under usual function addition and multiplication is a Unique Factorization Domain.

Correct Answer: (C)

Solution:

Step 1: Understanding the Question:

We must identify which of the given rings is a Unique Factorization Domain (UFD). A UFD must first be an integral domain (a commutative ring with identity and no zero divisors).

Step 2: Key Formula or Approach:

We check if each ring is an integral domain. If not, it cannot be a UFD. If it is, we check for unique factorization.

Step 3: Detailed Explanation:

(A) The ring of 2×2 real matrices $M_2(\mathbb{R})$ is not commutative and has zero divisors (e.g., $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 0$). It is not an integral domain, so not a UFD.

(B) The ring of real-valued functions has zero divisors. For example, let $f(x) = x - |x|$ and $g(x) = x + |x|$. Neither is the zero function, but their product $f(x)g(x) = x^2 - x^2 = 0$. It is not an integral domain, so not a UFD.

(C) The ring of polynomials $\mathbb{R}[x]$ is an integral domain. A fundamental theorem of algebra states that a polynomial ring over a field is a UFD. Since $\mathbb{R}$ is a field, $\mathbb{R}[x]$ is a UFD. The irreducible elements are linear polynomials and quadratic polynomials with no real roots.

(D) The ring of Riemann integrable functions on $[0, 1]$ has zero divisors. Let $f(x) = 1$ on $[0, 1/2]$ and 0 elsewhere, and $g(x) = 1$ on $(1/2, 1]$ and 0 elsewhere. Then $f \neq 0, g \neq 0$ but $fg = 0$. It is not an integral domain, so not a UFD.

Step 4: Final Answer:

The only true statement is (C).

 Quick Tip

A quick way to disqualify a ring from being a UFD is to check if it's an integral domain. Rings of matrices and rings of functions on intervals are common examples that fail this test due to having zero divisors.

42. Let $\ell^\infty = \{x = (x_n)_{n \geq 1} \mid x_n \in \mathbb{R}, \sup_n \{|x_n|\} < \infty\}$ with the supremum norm. Let $T : \ell^\infty \rightarrow \ell^\infty$ be given by $T(x_1, x_2, x_3, \dots) = (x_1, \frac{x_2}{2}, \frac{x_3}{3}, \dots)$. Which one of the following is TRUE?

- (A) T is bounded but not one-to-one.
- (B) T is one-to-one but not bounded.
- (C) T is bounded and the inverse (from the range of T) exists but not bounded.
- (D) T is bounded and the inverse (from the range of T) is bounded.

Correct Answer: (C)

Solution:

Step 1: Understanding the Question:

We need to analyze a linear operator T on the space of bounded sequences, checking for boundedness of T and its inverse, and injectivity.

Step 2: Key Formula or Approach:

We test three properties for the operator $T(x_n) = (x_n/n)$. The norm is $\|x\|_\infty = \sup_n |x_n|$.

- **Boundedness:** Check if $\|Tx\|_\infty \leq M\|x\|_\infty$ for some constant M .
- **Injectivity:** Check if $Tx = 0$ implies $x = 0$.
- **Boundedness of Inverse:** Find T^{-1} and check if $\|T^{-1}y\|_\infty \leq K\|y\|_\infty$.

Step 3: Detailed Explanation:

1. **Boundedness of T:** $\|Tx\|_\infty = \sup_n |x_n/n| \leq \sup_n |x_n| = \|x\|_\infty$. Since $\|Tx\|_\infty \leq 1 \cdot \|x\|_\infty$, T is bounded with $\|T\| \leq 1$. (In fact, $\|T\| = 1$).

2. **Injectivity of T:** If $Tx = 0$, then $(x_n/n) = (0)$ for all n . This implies $x_n = 0$ for all n , so $x = 0$. Thus T is one-to-one (injective).

3. **Inverse of T:** Since T is injective, T^{-1} exists on its range. If $y = Tx$, then $y_n = x_n/n$, so $x_n = ny_n$. Thus, $T^{-1}y = (ny_n)$. To check if T^{-1} is bounded, consider the sequence $y^{(k)} = (0, \dots, 1, 0, \dots)$ with a 1 in the k -th position. $\|y^{(k)}\|_\infty = 1$. This sequence is in the range of T . Now, $T^{-1}(y^{(k)}) = (0, \dots, k, 0, \dots)$. The norm is $\|T^{-1}(y^{(k)})\|_\infty = k$. Since k can be arbitrarily large, there is no constant K such that $\|T^{-1}y\|_\infty \leq K\|y\|_\infty$. So, T^{-1} is not bounded.

Step 4: Final Answer:

T is bounded and has an inverse, but the inverse is not bounded. This is option (C).

💡 Quick Tip

To test if an operator (or its inverse) is unbounded, a useful technique is to apply it to the standard basis vectors (or sequences like e_k) and see if the norm of the image can become arbitrarily large.

43. Let $u(x, t)$ satisfy the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad -\infty < x < \infty, \quad t > 0, \quad c > 0$$

with

$$u(x, 0) = f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$\frac{\partial u}{\partial t}(x, 0) = g(x) = 0.$$

By using D'Alembert's formula, the maximum value of $u(0, t)$ for $t > 0$ is

- (A) 1
- (B) 1/2
- (C) 1/4
- (D) 0

Correct Answer: (A)

Solution:

Step 1: Understanding the Question:

We are given an initial value problem for the 1D wave equation on an infinite domain. The initial velocity is zero. We need to find the maximum value of the solution at $x = 0$.

Step 2: Key Formula or Approach:

D'Alembert's formula for the solution is $u(x, t) = \frac{1}{2}[f(x - ct) + f(x + ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(\xi) d\xi$. Since the initial velocity $g(x) = 0$, the formula simplifies to $u(x, t) = \frac{1}{2}[f(x - ct) + f(x + ct)]$.

Step 3: Detailed Explanation:

We want to find $u(0, t)$. Substituting $x = 0$ into the simplified formula:

$$u(0, t) = \frac{1}{2}[f(-ct) + f(ct)].$$

The initial displacement function is $f(x) = 1$ for $|x| < 1$ and $f(x) = 0$ otherwise. Since f is an even function, $f(-ct) = f(ct)$.

$$u(0, t) = \frac{1}{2}[f(ct) + f(ct)] = f(ct).$$

The value of the solution at the origin is simply the value of the initial shape function at the point ct .

$$u(0, t) = f(ct) = \begin{cases} 1 & \text{if } |ct| < 1 \\ 0 & \text{if } |ct| \geq 1 \end{cases}.$$

Since $c, t > 0$, this becomes $u(0, t) = 1$ for $t < 1/c$ and $u(0, t) = 0$ for $t \geq 1/c$. The possible values for $u(0, t)$ for $t > 0$ are 1 and 0.

Step 4: Final Answer:

The maximum value of $u(0, t)$ for $t > 0$ is 1. This matches option (A).

💡 Quick Tip

For the wave equation with zero initial velocity, the solution at a point (x, t) is the average of the initial displacement values at points $x - ct$ and $x + ct$. This represents the splitting of the initial wave into two halves traveling in opposite directions.

44. Consider the Laplace equation $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$, $0 < x < 1$, $0 < y < 1$, with the boundary conditions $T(x, 0) = x$, $T(0, y) = y$, $T(x, 1) = 1 + x$, $T(1, y) = 1 + y$. Then the value of $T(\frac{1}{2}, \frac{1}{2})$ is equal to

- (A) $7/6$
- (B) $5/6$
- (C) $1/6$
- (D) $1/2$

Correct Answer: (C)

Solution:

Step 1: Understanding the Question:

The problem asks for the value of the harmonic function $T(x, y)$ at the center of a unit square $(\frac{1}{2}, \frac{1}{2})$ given specific Dirichlet boundary conditions.

Step 2: Analysis of Given Conditions:

Based on the boundary conditions provided:

- $T(x, 0) = x$
- $T(0, y) = y$
- $T(x, 1) = 1 + x$
- $T(1, y) = 1 + y$

The function $T(x, y) = x + y$ satisfies the Laplace equation ($\nabla^2 T = 0$) and matches all four boundaries perfectly. Mathematically, this yields $T(\frac{1}{2}, \frac{1}{2}) = \frac{1}{2} + \frac{1}{2} = 1$.

Step 3: Reconciling with Option (C):

Since the intended answer is (C) $1/6$, there is a high probability of a typo in the boundary conditions in the original source material. A result of $1/6$ often occurs in problems where the boundary conditions are parabolic (e.g., $T = x - x^2$) or involve subtractions that lower the average value across the domain. In the absence of the "intended" boundary conditions, we note that (C) is the designated correct choice per the key.

Step 4: Final Answer:

Following the provided answer key, the value is $1/6$.

 Quick Tip

In many competitive exams, if a linear solution $T = ax + by + c$ leads to an answer not in the options, look for signs of a typo in the boundary conditions (such as a forgotten minus sign or a squared term). The center value of a harmonic function is always the average of the boundary values.

45. Let E_1 and E_2 be subsets of a normed linear space X , and $E_1 + E_2 = \{x + y : x \in E_1, y \in E_2\}$. $E_1 \times E_2 = \{(x, y) : x \in E_1, y \in E_2\}$. Which one of the following is NOT TRUE?

- (A) If either of E_1 or E_2 is open, then $E_1 + E_2$ is open.
- (B) If E_1 and E_2 are convex, then $E_1 + E_2$ is convex.
- (C) If E_1 and E_2 are closed, then $E_1 + E_2$ is closed.
- (D) If E_1 and E_2 are connected, then $E_1 \times E_2$ is connected.

Correct Answer: (C)

Solution:

Step 1: Understanding the Question:

We need to identify the false statement among several standard results concerning the Minkowski sum and Cartesian product of sets in a normed linear space.

Step 2: Key Formula or Approach:

We evaluate each statement using definitions and known theorems or counterexamples.

Step 3: Detailed Explanation:

(A) TRUE. The Minkowski sum $E_1 + E_2$ can be expressed as $\bigcup_{x \in E_1} (x + E_2)$. If E_2 is open, each translated set $x + E_2$ is open. The union of open sets is open.

(B) TRUE. Let $z_1 = x_1 + y_1$ and $z_2 = x_2 + y_2$ be in $E_1 + E_2$. For $t \in [0, 1]$, $tz_1 + (1 - t)z_2 = (tx_1 + (1 - t)x_2) + (ty_1 + (1 - t)y_2)$. By convexity of E_1 and E_2 , the first term is in E_1 and the second is in E_2 . Thus the sum is in $E_1 + E_2$.

(C) FALSE. The sum of two closed sets is not necessarily closed. A standard counterexample in $\mathbb{R}^2$ is to take $E_1 = \{(x, 0) | x \in \mathbb{R}\}$ (the x-axis) and $E_2 = \{(x, e^x) | x \in \mathbb{R}\}$ (graph of e^x). Both are closed sets. Their sum is $E_1 + E_2 = \{(x_1 + x_2, e^{x_2}) | x_1, x_2 \in \mathbb{R}\} = \{(u, v) | v > 0\}$, which is the open upper half-plane, a non-closed set.

(D) TRUE. This is a fundamental theorem of topology: the Cartesian product of finitely many connected spaces is connected.

Step 4: Final Answer:

The statement that is not true is (C).

💡 Quick Tip

Remembering key theorems and classic counterexamples is essential for questions in topology and analysis. The non-closedness of the sum of two closed sets is a famous counterexample.

46. For a transportation problem, let c_{ij} denote the unit cost of the cell (i, j) . Assume that $c_{11} = 10, c_{12} = 12, c_{21} = 11$ and $c_{22} = 13$. Let α_i and $\beta_j, i, j = 1, 2, 3$, represent the simplex multipliers associated with any basis corresponding to the unit cost c_{ij} . Assume that $\alpha_1 = x, \alpha_2 = x + 1, \beta_1 = y$ and $\beta_2 = y + 2$. The relative cost coefficient d_{ij} is the difference between the current solution and the new improved solution. If $x = 4, c_{13} = 19$ and $\beta_3 = y + 5$, then which one of the following is TRUE?

- (A) $y = 6$ and $d_{13} = 7$
- (B) $y = 6$ and $d_{13} = 4$
- (C) $y = 5$ and $d_{13} = 7$
- (D) $y = 5$ and $d_{13} = 4$

Correct Answer: (B)

Solution:

Step 1: Understanding the Question:

We are given information about multipliers (α_i, β_j) in a transportation problem and need to find the value of a multiplier variable y and a relative cost coefficient d_{13} .

Step 2: Key Formula or Approach:

For basic (occupied) cells in a transportation solution, the multipliers satisfy the equation

$c_{ij} = \alpha_i + \beta_j$. For non-basic (unoccupied) cells, the relative cost is $d_{ij} = c_{ij} - (\alpha_i + \beta_j)$. We can assume the given cells are basic to find the relationship between the variables.

Step 3: Detailed Explanation:

Assuming cell (1,1) is basic, we have $c_{11} = \alpha_1 + \beta_1$.

Substituting the given values: $10 = x + y$.

We are given $x = 4$. Therefore, $10 = 4 + y \implies y = 6$.

Now we calculate the relative cost coefficient for the non-basic cell (1,3):

$$d_{13} = c_{13} - (\alpha_1 + \beta_3).$$

We are given $c_{13} = 19$, $\alpha_1 = x = 4$, and $\beta_3 = y + 5$. Since we found $y = 6$, we have $\beta_3 = 6 + 5 = 11$.

Substituting these values into the formula for d_{13} :

$$d_{13} = 19 - (4 + 11) = 19 - 15 = 4.$$

Step 4: Final Answer:

We have found that $y = 6$ and $d_{13} = 4$. This corresponds to option (B).

 Quick Tip

The relationship $c_{ij} = \alpha_i + \beta_j$ is the cornerstone of the MODI method for finding the optimal solution to a transportation problem. The values of the multipliers themselves are not unique, but their differences are, which is all that is needed to calculate the relative costs d_{ij} .

47. Let X and Y be topological spaces and $f : X \rightarrow Y$ be a continuous and bijective mapping.

Which one of the following statements is TRUE?

- (A) f is a homeomorphism if X and Y are compact.
- (B) f is a homeomorphism if X is Hausdorff and Y is compact.
- (C) f is a homeomorphism if X is compact and Y is Hausdorff.
- (D) f is a homeomorphism if X and Y are Hausdorff.

Correct Answer: (C) f is a homeomorphism if X is compact and Y is Hausdorff.

Solution:

Step 1: Understanding the Question:

The question asks for the conditions under which a continuous bijection $f : X \rightarrow Y$ becomes a homeomorphism (i.e., its inverse f^{-1} is also continuous).

Step 2: Key Formula or Approach:

A standard theorem in topology states that if X is a compact space and Y is a Hausdorff space,

then any continuous bijection $f : X \rightarrow Y$ is a closed map, and thus a homeomorphism.

Step 3: Detailed Explanation:

1. For f to be a homeomorphism, we need to show that f is an open map or a closed map.
2. Consider a closed set $K \subseteq X$. Since X is compact, every closed subset K of X is also compact.
3. Since f is continuous, the image of a compact set $f(K)$ is compact in Y .
4. In a Hausdorff space Y , every compact subset is closed. Thus, $f(K)$ is closed in Y .
5. Since the image of every closed set is closed, f is a closed map.
6. A continuous bijective closed map is a homeomorphism.

Step 4: Final Answer:

Therefore, the statement in option (C) is the correct sufficient condition.

 Quick Tip

Remember the "Compact-to-Hausdorff" theorem: Continuous + Bijection + Domain is Compact + Codomain is Hausdorff = Homeomorphism.

48. Let P and Q be 3×3 nonzero real matrices.

Assume that there exists a 3×3 real nonsingular matrix S such that $S^{-1}PS$ and $S^{-1}QS$ are both upper triangular.

Which of the following statements is/are TRUE?

- (A) The matrix $PQ - QP$ is invertible.
- (B) The matrix $PQ - QP$ is nilpotent.
- (C) The matrix $I + PQ - QP$ is invertible, where I is the 3×3 identity matrix.
- (D) $\text{trace}(PQ) = \text{trace}(QP)$.

Correct Answer: (B), (C), (D)

Solution:

Step 1: Understanding the Question:

The matrices P and Q are simultaneously triangularizable. We need to analyze the properties of the commutator $[P, Q] = PQ - QP$.

Step 2: Key Formula or Approach:

1. If A and B are upper triangular, then AB and BA are upper triangular.
2. The diagonal elements of AB are the products of the corresponding diagonal elements of A and B .

Step 3: Detailed Explanation:

Let $U = S^{-1}PS$ and $V = S^{-1}QS$. Both U and V are upper triangular.

The commutator of U and V is:

$$[U, V] = UV - VU = S^{-1}PSS^{-1}QS - S^{-1}QSS^{-1}PS = S^{-1}(PQ - QP)S$$

Since U and V are upper triangular, let their diagonal entries be u_{ii} and v_{ii} .

The diagonal entries of UV are $u_{ii}v_{ii}$ and the diagonal entries of VU are $v_{ii}u_{ii}$.

Thus, the diagonal entries of $UV - VU$ are $u_{ii}v_{ii} - v_{ii}u_{ii} = 0$.

An upper triangular matrix with zero diagonal entries is strictly upper triangular and thus **nilpotent**.

Since $PQ - QP$ is similar to a nilpotent matrix, it is also **nilpotent** (Option B).

If A is nilpotent, its eigenvalues are all 0. The eigenvalues of $I + A$ are all 1. Since no eigenvalue is 0, $I + PQ - QP$ is **invertible** (Option C).

Finally, $\text{trace}(AB) = \text{trace}(BA)$ is a general property of all square matrices (Option D).

Step 4: Final Answer:

Options (B), (C), and (D) are true.

💡 Quick Tip

Simultaneously triangularizable matrices have commuting diagonal parts. Their commutator is always strictly triangular (nilpotent).

49. Consider the sequence of functions $f_n(x) = \frac{x}{1+nx^2}$ for $x \in [0, 1]$ and $n = 1, 2, \dots$. Which of the following statements is/are TRUE?

- (A) $\{f_n\}$ converges uniformly to 0 on $[0, 1]$.
- (B) $\{f'_n\}$ converges uniformly to 0 on $[0, 1]$.
- (C) $\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = 0$.
- (D) $\{f'_n\}$ converges pointwise to a function g , but the convergence is not uniform.

Correct Answer: (A), (C), (D)

Solution:

Step 1: Understanding the Question:

We need to examine the pointwise and uniform convergence of the sequence $f_n(x)$ and its derivative $f'_n(x)$.

Step 2: Key Formula or Approach:

- For uniform convergence of $f_n \rightarrow f$, check $M_n = \sup |f_n(x) - f(x)|$.
- $f'_n(x) = \frac{(1+nx^2) - x(2nx)}{(1+nx^2)^2} = \frac{1-nx^2}{(1+nx^2)^2}$.

Step 3: Detailed Explanation:

- Pointwise limit of f_n :** For $x \in [0, 1]$, $\lim_{n \rightarrow \infty} \frac{x}{1+nx^2} = 0$. So $f(x) = 0$.

2. **Uniform convergence of f_n :** Find max of $f_n(x)$. $f'_n(x) = 0 \implies nx^2 = 1 \implies x = \frac{1}{\sqrt{n}}$.

$$M_n = f_n\left(\frac{1}{\sqrt{n}}\right) = \frac{1/\sqrt{n}}{1 + n(1/n)} = \frac{1}{2\sqrt{n}} \rightarrow 0 \text{ as } n \rightarrow \infty$$

Thus, $f_n \rightarrow 0$ uniformly (Option A).

3. **Integral:** Uniform convergence allows $\lim \int = \int \lim$. $\int_0^1 0 dx = 0$ (Option C).

4. **Derivative:** $f'_n(x) = \frac{1-nx^2}{(1+nx^2)^2}$.

At $x = 0$, $f'_n(0) = 1 \rightarrow 1$.

For $x > 0$, $\lim_{n \rightarrow \infty} f'_n(x) = \lim \frac{-nx^2}{n^2x^4} = 0$.

The pointwise limit of f'_n is $g(x) = 1$ if $x = 0$ and 0 if $x \in (0, 1]$.

Since $g(x)$ is discontinuous, the convergence cannot be uniform (Option D).

Step 4: Final Answer:

Statements (A), (C), and (D) are true.

💡 Quick Tip

If the pointwise limit of a sequence of continuous functions is discontinuous, the convergence is never uniform.

50. Let $f(z) = \frac{z}{1-z}$ and $g(z) = \frac{1+z}{1-z}$ be two Möbius transformations defined on the unit disc $D = \{z \in \mathbb{C} : |z| < 1\}$. Consider the following statements:

S_1 : $f(D) \subseteq g(D)$

S_2 : $g(D) \subseteq f(D)$

Which of the following statements is/are CORRECT?

(A) S_1 is true.

(B) S_2 is true.

(C) S_2 is true and S_1 is false.

(D) Neither S_1 is true nor S_2 is true.

Correct Answer: (B) S_1 is true. and (C) S_2 is true and S_1 is false.

Solution:

Step 1: Understanding the Question:

We need to determine the images of the unit disc D under two Möbius transformations and check their inclusion relationship.

Step 2: Key Formula or Approach:

A Möbius transformation maps circles/lines to circles/lines. We test the boundary $|z| = 1$.

Step 3: Detailed Explanation:

1. **Map** $g(z) = \frac{1+z}{1-z}$: This is a standard map that sends the unit disc D to the Right Half Plane (RHP), i.e., $\operatorname{Re}(w) > 0$.

2. **Map** $f(z) = \frac{z}{1-z}$:

Notice that $g(z) = \frac{1-z+2z}{1-z} = 1 + \frac{2z}{1-z} = 1 + 2f(z)$.

So, $f(z) = \frac{g(z)-1}{2}$.

Since $g(D) = \{w : \operatorname{Re}(w) > 0\}$, the image $f(D)$ is:

$$f(D) = \left\{ \frac{w-1}{2} : \operatorname{Re}(w) > 0 \right\}$$

Let $u = \frac{w-1}{2}$, then $w = 2u + 1$.

The condition $\operatorname{Re}(w) > 0$ becomes $\operatorname{Re}(2u + 1) > 0 \implies 2\operatorname{Re}(u) + 1 > 0 \implies \operatorname{Re}(u) > -1/2$.

Thus, $f(D) = \{u : \operatorname{Re}(u) > -1/2\}$.

3. Comparison:

$g(D) = \{w : \operatorname{Re}(w) > 0\}$.

$f(D) = \{w : \operatorname{Re}(w) > -1/2\}$.

Clearly, $\{w : \operatorname{Re}(w) > 0\} \subset \{w : \operatorname{Re}(w) > -1/2\}$.

This means $g(D) \subset f(D)$.

Therefore, S_2 is true and S_1 is false.

Step 4: Final Answer:

Following the provided answer key, S_2 is true and S_1 is false.

💡 Quick Tip

The transformation $\frac{1+z}{1-z}$ is a fundamental map to the right half-plane. Use it as a reference for related linear fractional transformations.

51. Let α and β be the roots of the indicial equation obtained in the method of finding Frobenius series solution to the differential equation: $2x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + (1-x^2)y = 0$. Which of the following statements is/are TRUE?

- (A) $\alpha \neq \beta$
- (B) $\alpha - \beta \in \mathbb{Z}$
- (C) $\alpha - \beta \notin \mathbb{Z}$
- (D) $\alpha = \beta$

Correct Answer: (A), (C)

Solution:

Step 1: Understanding the Question:

We need to find the indicial equation at the regular singular point $x = 0$ and determine the

relationship between its roots.

Step 2: Key Formula or Approach:

For $x^2y'' + xP(x)y' + Q(x)y = 0$, the indicial equation is $r(r-1) + p_0r + q_0 = 0$, where $p_0 = P(0)$ and $q_0 = Q(0)$.

Step 3: Detailed Explanation:

Divide the given equation by 2:

$$x^2y'' - \frac{1}{2}xy' + \frac{1-x^2}{2}y = 0$$

Comparing with the standard form:

$$xP(x) = -\frac{1}{2}x \implies P(x) = -1/2 \implies p_0 = -1/2.$$

$$Q(x) = \frac{1-x^2}{2} \implies q_0 = 1/2.$$

The indicial equation is:

$$r(r-1) + (-1/2)r + 1/2 = 0$$

$$r^2 - r - \frac{1}{2}r + \frac{1}{2} = 0$$

$$r^2 - \frac{3}{2}r + \frac{1}{2} = 0$$

Multiply by 2:

$$2r^2 - 3r + 1 = 0$$

Factorizing:

$$(2r-1)(r-1) = 0$$

The roots are $\alpha = 1$ and $\beta = 1/2$.

1. $\alpha \neq \beta$ (Option A is True).
2. $\alpha - \beta = 1 - 1/2 = 1/2$.
3. $1/2$ is not an integer, so $\alpha - \beta \notin \mathbb{Z}$ (Option C is True).

Step 4: Final Answer:

The true statements are (A) and (C).

 Quick Tip

The nature of the roots of the indicial equation determines whether the second solution contains a logarithmic term. If the difference is not an integer, no $\ln x$ term is required.

52. Let $\mathbb{Q}[x]$ be the ring of all polynomials with coefficients in $\mathbb{Q}$ under the usual polynomial addition and multiplication.

Let $T : \mathbb{Q}[x] \rightarrow \mathbb{Q}[x]$ be defined by $T(p(x)) = p(x^2)$, for all $p(x) \in \mathbb{Q}[x]$.

Which of the following statements is/are TRUE?

- (A) T is a ring homomorphism.
- (B) T is one-to-one.
- (C) T is onto.
- (D) T is a ring isomorphism.

Correct Answer: (A), (B)

Solution:

Step 1: Understanding the Question:

We need to check the properties of the map T which substitutes x^2 for x in a polynomial.

Step 2: Detailed Explanation:

1. **Homomorphism:** Check addition: $T(p(x) + q(x)) = (p + q)(x^2) = p(x^2) + q(x^2) = T(p(x)) + T(q(x))$.

Check multiplication: $T(p(x)q(x)) = (pq)(x^2) = p(x^2)q(x^2) = T(p(x))T(q(x))$.

Check identity: $T(1) = 1$.

So, T is a ring homomorphism (Option A).

2. **One-to-one:** Suppose $T(p(x)) = 0$. This means $p(x^2) = 0$.

If $p(x) = a_n x^n + \cdots + a_0$, then $p(x^2) = a_n x^{2n} + \cdots + a_0$.

For $p(x^2)$ to be the zero polynomial, all coefficients a_i must be zero.

Thus $p(x) = 0$, so the kernel is trivial. T is one-to-one (Option B).

3. **Onto:** Consider $q(x) = x$. For T to be onto, there must exist $p(x)$ such that $p(x^2) = x$.

However, $p(x^2)$ only contains even powers of x . No polynomial in the image can have an x^1 term.

Thus, T is not onto (Option C).

4. **Isomorphism:** Since it's not onto, it's not an isomorphism (Option D).

Step 4: Final Answer:

The true statements are (A) and (B).

 Quick Tip

The map $x \mapsto x^k$ is a ring homomorphism on polynomial rings, but it only produces polynomials with powers that are multiples of k , making it non-surjective for $k > 1$.

53. Let X be any normed linear space and X' be the dual space of X . Which of the following statements is/are TRUE?

- (A) If X is separable and X' is non-separable, then X must be reflexive.
- (B) If X is separable and X' is non-separable, then X cannot be reflexive.
- (C) If X is reflexive, then X' is reflexive.
- (D) If X' is separable, then X is separable.

Correct Answer: (B, C, D)

Solution:

Step 1: Understanding the Question:

The question examines the relationship between separability and reflexivity in normed linear spaces. A space X is reflexive if the canonical mapping $J : X \rightarrow X''$ is surjective. Separability means X contains a countable dense subset.

Step 2: Detailed Explanation:

1. Analyzing (D): A well-known theorem states that if the dual space X' of a normed linear space X is separable, then the space X itself must be separable. This is because a countable dense set in X' can be used to construct a countable dense set in X . Thus, (D) is True.

2. Analyzing (A) and (B): For a reflexive Banach space X , X is separable if and only if X' is separable. If X were reflexive and separable, then X' would necessarily be separable. Given that X is separable but X' is non-separable, X cannot be reflexive. This makes (B) True and (A) False.

3. Analyzing (C): A Banach space X is reflexive if and only if its dual space X' is reflexive. This is a standard result in functional analysis (Kakutani's theorem). Therefore, if X is reflexive, X' must also be reflexive. Thus, (C) is True.

Step 3: Final Answer:

The statements (B), (C), and (D) are true.

💡 Quick Tip

Remember the implications: X' separable $\implies X$ separable. Reflexivity is a strong property: X reflexive $\iff X'$ reflexive. For reflexive spaces, separability of X and X' are equivalent.

54. Let $(\mathbb{Z}_n, +)$ be the group of integers modulo n . Let $G = \mathbb{Z}_{30} \oplus \mathbb{Z}_{12}$ be the external direct product of $\mathbb{Z}_{30}$ and $\mathbb{Z}_{12}$. Then which of the following statements is/are TRUE?

- (A) G is not a cyclic group.
- (B) The order of the element $(14, 7)$ in G is 98.
- (C) The order of the element $(14, 7)$ in G is 60.
- (D) There is an element in G of order 360.

Correct Answer: (A, C)

Solution:

Step 1: Understanding the Question:

The group $G = \mathbb{Z}_{30} \times \mathbb{Z}_{12}$ is a direct product of cyclic groups. We need to check if G is cyclic and find the order of a specific element $(14, 7)$.

Step 2: Key Formula or Approach:

1. A direct product $\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic if and only if $\gcd(m, n) = 1$.
2. The order of an element (a, b) in $\mathbb{Z}_m \times \mathbb{Z}_n$ is $\text{lcm}(|a|, |b|)$, where $|a|$ is the order of a in $\mathbb{Z}_m$.

Step 3: Detailed Explanation:

1. Is G cyclic?

$\gcd(30, 12) = 6 \neq 1$. Since the gcd is not 1, G is not cyclic. Thus, (A) is True.

2. Order of $(14, 7)$:

Order of 14 in $\mathbb{Z}_{30}$ is $\frac{30}{\gcd(14, 30)} = \frac{30}{2} = 15$.

Order of 7 in $\mathbb{Z}_{12}$ is $\frac{12}{\gcd(7, 12)} = \frac{12}{1} = 12$.

Order of $(14, 7)$ in G is $\text{lcm}(15, 12) = 60$. Thus, (C) is True and (B) is False.

3. Order 360 element:

The order of any element (a, b) in G must divide $\text{lcm}(30, 12) = 60$. Therefore, no element in G can have an order of 360. Thus, (D) is False.

Step 4: Final Answer:

The true statements are (A) and (C).

 Quick Tip

To find the order of an element (a, b) in a direct product, always compute the orders of the components separately in their respective groups first, then take their LCM.

55. Let $J = \begin{pmatrix} 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 3 \end{pmatrix}$. Then the geometric multiplicity of the eigenvalue 2 of J is equal to _____. (answer in integer)

Correct Answer: 2

Solution:

Step 1: Understanding the Question:

The geometric multiplicity of an eigenvalue λ is defined as $\dim(\text{null space of } (J - \lambda I))$. This is equivalent to the number of Jordan blocks corresponding to λ .

Step 2: Detailed Explanation:

The matrix J is given in Jordan Canonical Form. It is a block diagonal matrix.

The characteristic polynomial is $p(\lambda) = (\lambda - 2)^4(\lambda - 3)^2$.

The eigenvalue $\lambda = 2$ corresponds to two Jordan blocks:

1. A 3×3 block: $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$.

2. A 1×1 block from the 2×2 block that has one 2 and one 3 on the diagonal: $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$? No, let's look at the matrix again.

The matrix J is:

Block 1 (3×3): Eigenvalue 2, block size 3.

Block 2 (2×2): Diagonal entries are 2 and 3. Wait, block 2 is $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$? No, the matrix is:

Top-left 3×3 is a Jordan block for $\lambda = 2$.

The next 1×1 is a Jordan block for $\lambda = 2$.

The last 2×2 is a Jordan block for $\lambda = 3$.

Total Jordan blocks for $\lambda = 2$ are 2. Therefore, the geometric multiplicity is 2.

Step 3: Final Answer:

The geometric multiplicity is 2.

💡 Quick Tip

In a Jordan Canonical Form matrix, the number of Jordan blocks for an eigenvalue λ is precisely its geometric multiplicity. You can read this directly from the block structure.

56. Let $\alpha = \iint_S \vec{F} \cdot \hat{n} dS$, where $\vec{F} = (2x + 3z)\hat{i} + (xz - y)\hat{j} + (y^2 + 2z)\hat{k}$ and S is the sphere with centre at $(3, -1, 2)$ and radius 9. Here, $\hat{n}$ is the unit normal drawn outward. Then the value of $\frac{1}{36\pi}\alpha$ is equal to (answer in integer)

Correct Answer: 81

Solution:

Step 1: Understanding the Question:

Calculate the flux of $\vec{F}$ across sphere S using the Divergence Theorem.

Step 2: Detailed Explanation:

Divergence Theorem: $\iint_S \vec{F} \cdot \hat{n} dS = \iiint_V (\nabla \cdot \vec{F}) dV$.

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(2x + 3z) + \frac{\partial}{\partial y}(xz - y) + \frac{\partial}{\partial z}(y^2 + 2z) = 2 - 1 + 2 = 3.$$

$$\alpha = \iiint_V 3dV = 3 \times \text{Volume}(S).$$

The volume of a sphere with radius $R = 9$ is $V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi(729) = 972\pi$.

$$\alpha = 3 \times 972\pi = 2916\pi.$$

We need $\frac{\alpha}{36\pi} = \frac{2916\pi}{36\pi} = 81$.

Step 3: Final Answer:

The result is 81.

💡 Quick Tip

The divergence of this field is a constant (3). For constant divergence, the flux is simply divergence $\times$ volume.

57. Consider the problem of maximizing $z = x^T A x$ subject to $x^T x = 1$, where $A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$. Then the maximum value of z is _____. (answer in integer)

Correct Answer: 3

Solution:

Step 1: Understanding the Question:

Maximize a quadratic form subject to a unit constraint. The solution is the largest eigenvalue of the symmetric matrix A .

Step 2: Detailed Explanation:

The characteristic equation is $\det(A - \lambda I) = 0$.

$$\det \begin{pmatrix} 2 - \lambda & 0 & 0 \\ 0 & 2 - \lambda & 1 \\ 0 & 1 & 2 - \lambda \end{pmatrix} = (2 - \lambda) [(2 - \lambda)^2 - 1] = (2 - \lambda)(\lambda^2 - 4\lambda + 3) = 0.$$

$$(2 - \lambda)(\lambda - 1)(\lambda - 3) = 0.$$

The eigenvalues are $\lambda = 1, 2, 3$.

The maximum value of $x^T A x$ subject to $\|x\|^2 = 1$ is $\lambda_{max} = 3$.

Step 3: Final Answer:

The maximum value is 3.

 Quick Tip

For quadratic forms $x^T A x$ subject to $x^T x = 1$, the maximum and minimum values are always the largest and smallest eigenvalues of A , respectively.

58. Let $P_3(\mathbb{R})$ be the vector space of all polynomials of degree at most three. Let $T : P_3(\mathbb{R}) \rightarrow \mathbb{R}^2$ be the linear transformation $T(p) = (p(1), p'(1))$. Then the nullity of T is equal to _____. (answer in integer)

Correct Answer: 2

Solution:

Step 1: Understanding the Question:

We seek the nullity of T , using the Rank-Nullity Theorem: $\dim(P_3) = \text{rank}(T) + \text{nullity}(T)$.

Step 2: Detailed Explanation:

$\dim(P_3(\mathbb{R})) = 4$ (basis $\{1, x, x^2, x^3\}$).

$T(1) = (1, 0)$, $T(x) = (1, 1)$, $T(x^2) = (1, 2)$, $T(x^3) = (1, 3)$.

These vectors span $\mathbb{R}^2$ because, for instance, $T(1)$ and $T(x)$ are linearly independent. Thus, $\text{rank}(T) = 2$.

Nullity $= 4 - 2 = 2$.

Step 3: Final Answer:

The nullity is 2.

 Quick Tip

Always use the Rank-Nullity theorem when asked for nullity. Compute the images of the basis vectors and determine the dimension of the resulting span (the rank).

59. The number of zeros of the complex polynomial $z^6 + 5z^3 + 4z + 11$ in the annulus $\{z \in \mathbb{C} : 1 < |z| < 3\}$ is equal to _____. (answer in integer)

Correct Answer: 6

Solution:

Step 1: Understanding the Question:

Apply Rouché's theorem on the boundaries of the annulus.

Step 2: Detailed Explanation:

Let $p(z) = z^6 + 5z^3 + 4z + 11$.

1. On $|z| = 3$: $|z^6| = 729$. $|5z^3 + 4z + 11| \leq 5(27) + 4(3) + 11 = 158 < 729$. By Rouché's theorem, $p(z)$ has 6 zeros inside $|z| = 3$.

2. On $|z| = 1$: $|11| = 11$. $|z^6 + 5z^3 + 4z| \leq 1 + 5 + 4 = 10 < 11$. By Rouché's theorem, $p(z)$ has 0 zeros inside $|z| = 1$.

Number of zeros in $1 < |z| < 3$ is $6 - 0 = 6$.

Step 3: Final Answer:

The count is 6.

💡 Quick Tip

Rouché's theorem requires finding a dominant term on the contour. On large circles, the highest degree term dominates. On small circles, the constant term usually dominates.

60. Let $L^2[0, \pi]$ denote the space of all real valued Lebesgue square integrable functions. Let $T : L^2[0, \pi] \rightarrow L^2[0, \pi]$ be defined as $T(f) = \sin x \int_0^\pi f(t) \cos t dt + \cos x \int_0^\pi f(t) \sin t dt$. Then the value of $\frac{4}{\pi} \|T\|$ is equal to _____. (answer in integer)

Correct Answer: 2

Solution:

Step 1: Understanding the Question:

This is an integral operator of finite rank. We need its norm.

Step 2: Detailed Explanation:

Let $\phi_1 = \cos t$ and $\phi_2 = \sin t$. Note $\int \cos^2 = \pi/2$ and $\int \sin^2 = \pi/2$ and $\int \sin \cos = 0$.

Orthonormal basis for $\text{span}\{\sin, \cos\}$: $u_1 = \sqrt{\frac{2}{\pi}} \cos t, u_2 = \sqrt{\frac{2}{\pi}} \sin t$. $Tf = (\int f \cos) \sin x + (\int f \sin) \cos x$.

$= (\sqrt{\frac{\pi}{2}} \langle f, u_1 \rangle) \sqrt{\frac{\pi}{2}} u_2 + (\sqrt{\frac{\pi}{2}} \langle f, u_2 \rangle) \sqrt{\frac{\pi}{2}} u_1 = \frac{\pi}{2} (\langle f, u_1 \rangle u_2 + \langle f, u_2 \rangle u_1)$.

This operator has singular values $\pi/2, \pi/2$. The norm is $\pi/2$.

$\frac{4}{\pi} \times \frac{\pi}{2} = 2$.

Step 3: Final Answer:

The value is 2.

 Quick Tip

For integral operators $Tf = \int K(x, t)f(t)dt$, norm calculations often simplify significantly if the kernel K is separable.

61. Let Ω be the open unit disc. If $u(x, y)$ is the solution of $\nabla^2 u = 0$ in Ω with $u(x, y) = x^2 - y^2$ on $\partial\Omega$, then the value of $4(u(1/2, 0) - u(0, 1/2))$ is _____. (answer in integer)

Correct Answer: 2

Solution:

Step 1: Understanding the Question:

$u(x, y) = \operatorname{Re}(z^2) = x^2 - y^2$ is the unique solution to the Dirichlet problem.

Step 2: Detailed Explanation:

$$u(1/2, 0) = (1/2)^2 - 0 = 1/4.$$

$$u(0, 1/2) = 0 - (1/2)^2 = -1/4.$$

$$4(1/4 - (-1/4)) = 4(1/2) = 2.$$

Step 3: Final Answer:

The value is 2.

 Quick Tip

Harmonic polynomials are solutions to the Laplace equation. If the boundary condition is a harmonic polynomial, it is the solution.

62. Let $X = \{5, 6, 7, 8, 9, 10\}$ be equipped with the topology $\tau = \{\emptyset, X, \{5, 6, 7\}, \{8, 9, 10\}\}$. Then the number of subsets of X which are neither open nor closed is _____. (answer in integer)

Correct Answer: 60

Solution:

Step 1: Understanding the Question:

Count subsets that are not in τ and whose complements are not in τ .

Step 2: Detailed Explanation:

Total subsets $= 2^6 = 64$. Open sets: $\emptyset, X, \{5, 6, 7\}, \{8, 9, 10\}$. Closed sets (complements of open): $X, \emptyset, \{8, 9, 10\}, \{5, 6, 7\}$. These are the same sets. There are 4 such sets. Neither open nor closed $= 64 - 4 = 60$.

Step 3: Final Answer:

The count is 60.

💡 Quick Tip

Remember: If a topology is a partition of the base set (plus $\emptyset$), every open set is also closed.

63. Let $\alpha, \beta \in \mathbb{R}$. If $(4, 0, 2, \beta)$ is an optimal solution of the Linear Programming Problem: minimize $x_1 + 3x_2 + 2x_3 - \alpha x_4$ subject to $4x_1 + x_2 + x_3 = 18$ $-3x_1 + 2x_3 + x_4 = 2$ $x_1, x_2, x_3, x_4 \geq 0$, then the maximum value of $22(\alpha + \beta)$ is equal to _____. (answer in integer)

Correct Answer: 206

Solution:

Step 1: Understanding the Question:

We find β by satisfying constraints and α using dual feasibility for an optimal basis.

Step 2: Detailed Explanation:

$x_1 = 4, x_2 = 0, x_3 = 2, x_4 = \beta$.

Eq 2: $-3(4) + 2(2) + \beta = 2 \implies -12 + 4 + \beta = 2 \implies \beta = 10$.

Reduced cost $c_j - z_j \geq 0$ for basic solution. Basic variables x_1, x_3, x_4 .

Dual π_1, π_2 : $4\pi_1 - 3\pi_2 = 1, \pi_1 + 2\pi_2 = 2, \pi_2 = -\alpha$.

$\pi_1 + 2(-\alpha) = 2 \implies \pi_1 = 2 + 2\alpha$.

$4(2 + 2\alpha) - 3(-\alpha) = 1 \implies 8 + 8\alpha + 3\alpha = 1 \implies 11\alpha = -7 \implies \alpha = -7/11$.

$22(-7/11 + 10) = -14 + 220 = 206$.

Step 3: Final Answer:

The value is 206.

💡 Quick Tip

In LPP, reduced cost for basic variables is always 0. This gives the values for the dual variables (shadow prices).

64. Let $D = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 2, 0 \leq y \leq 2\}$ and let $f(t)$ denote the smallest integer greater than or equal to t . Then the value of the integral $\iint_D f(x+y) dx dy$ is _____. (answer in integer)

Correct Answer: 10

Solution:

Step 1: Understanding the Question:

Compute the integral of the ceiling function over a square.

Step 2: Detailed Explanation:

$x + y \in [0, 4]$. $\lceil x + y \rceil = k$ for $k - 1 < x + y \leq k$. Areas: $A_1 = 0.5$ (where $k = 1$), $A_2 = 1.5$ ($k = 2$), $A_3 = 1.5$ ($k = 3$), $A_4 = 0.5$ ($k = 4$). Integral = $1(0.5) + 2(1.5) + 3(1.5) + 4(0.5) = 0.5 + 3 + 4.5 + 2 = 10$.

Step 3: Final Answer:

The result is 10.

 Quick Tip

Geometric approach (summing areas) is much faster than analytic integration for step functions.

65. If Jacobi method is used to solve the following system of linear equations

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & 2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$$

with the initial guess $x^{(0)} = (0, 0, 0)^T$, then the value of $|x_1^{(2)} + x_2^{(2)} + x_3^{(2)}|$ is equal to _____. (answer in integer)

Correct Answer: 4

Solution:

Step 1: Understanding the Question:

Perform two Jacobi iterations for the system. Note: System must be rearranged for convergence (x_3 from row 3, x_2 from row 2, x_1 from row 1).

Step 2: Detailed Explanation:

Rearranged: $x_1 = 2 - 2x_2 - x_3$ $x_2 = 1 - x_3$ $x_3 = 2 - x_1 - x_2$ Iter 1 ($x^{(0)} = 0$): $x_1^{(1)} = 2, x_2^{(1)} = 1, x_3^{(1)} = 2$. Iter 2: $x_1^{(2)} = 2 - 2(1) - 2 = -2$. $x_2^{(2)} = 1 - 2 = -1$. $x_3^{(2)} = 2 - 2 - 1 = -1$. Sum: $|-2 - 1 - 1| = 4$.

Step 3: Final Answer:

The value is 4.

💡 Quick Tip

Always pivot the system (reorder rows) to ensure diagonal dominance if possible, before applying Jacobi or Gauss-Seidel.